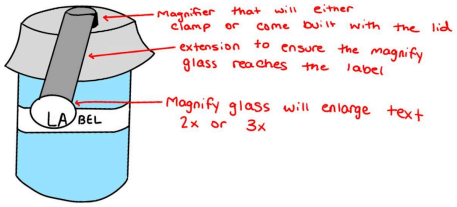
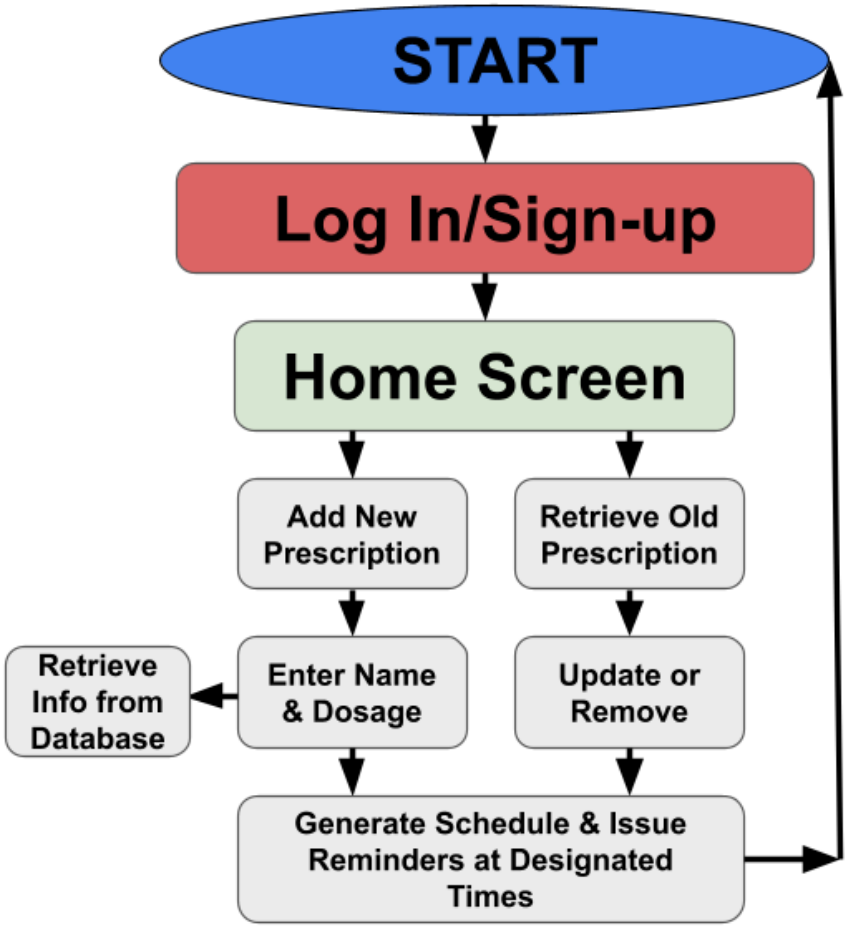
Element D: Design Concepts Generation, Analysis, and Selection

Concept Development / Brainstorming

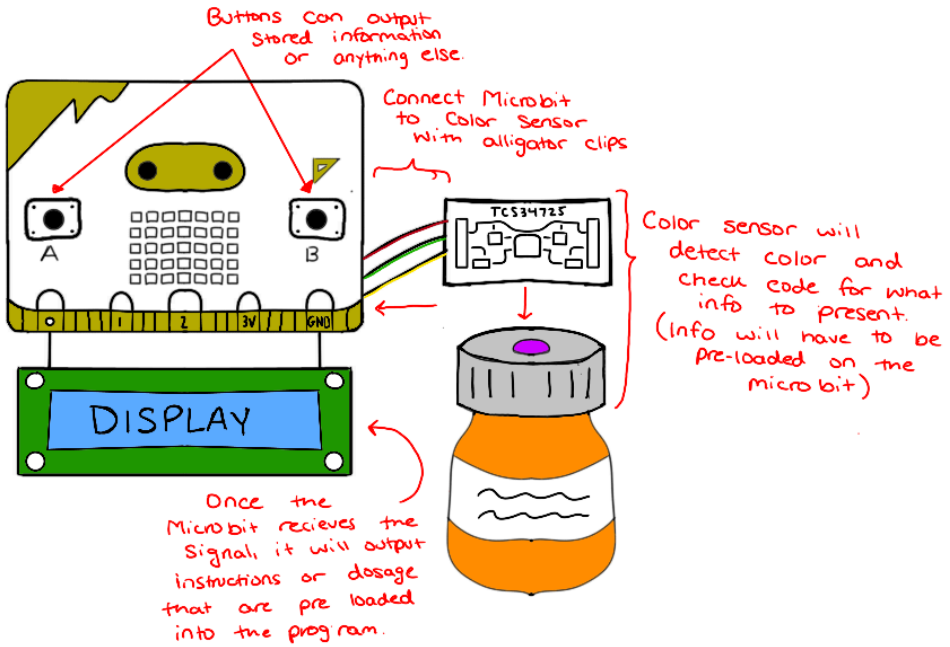


Clip-On Hinged Magnifier Lid
Prescription Readability
Dhruv Shah Brainstorming #1

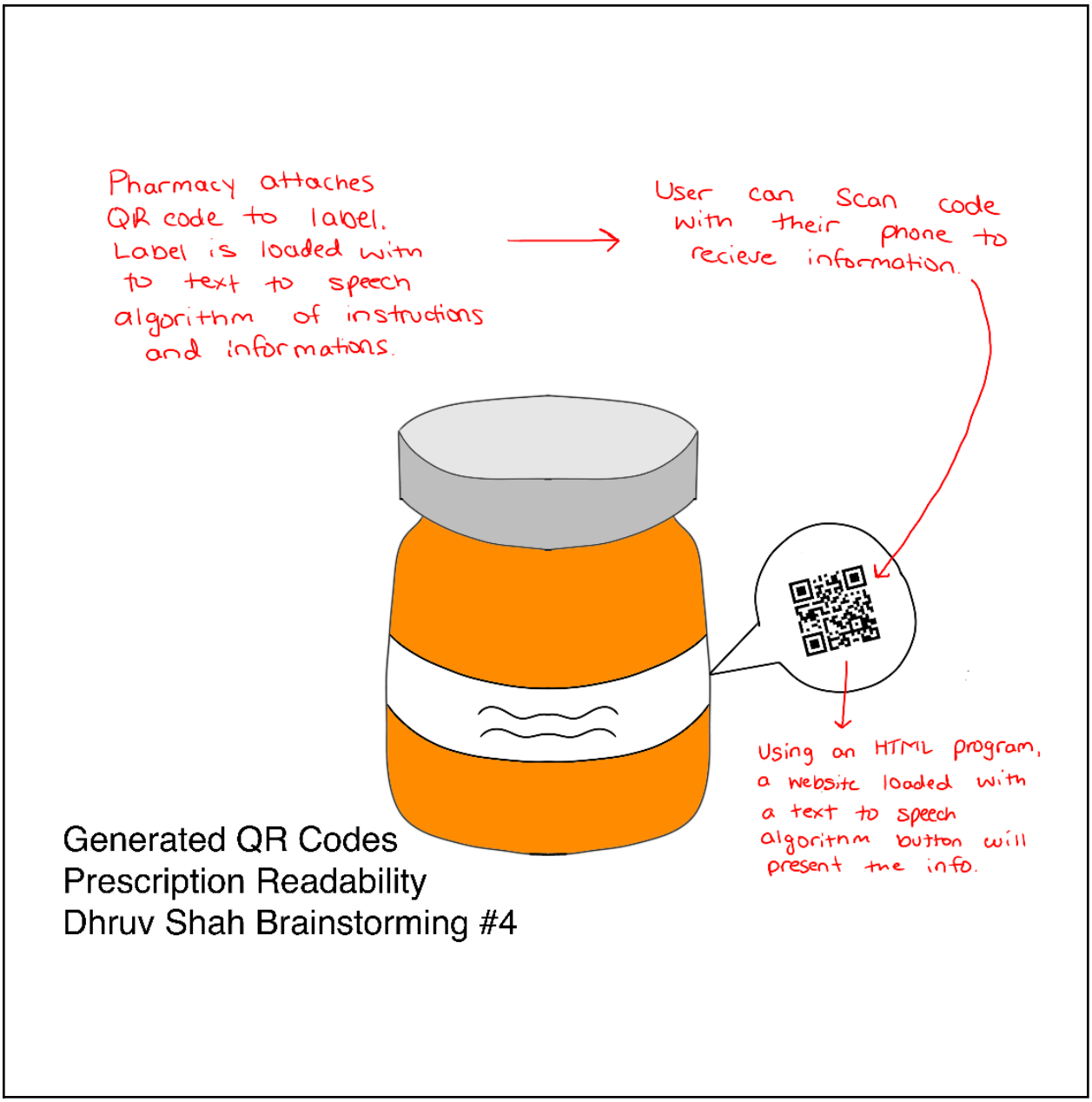
The Saturn App allows users to make calendars for high school students using a database of 17000 schools to organize classes. Similarly users can choose their prescriptions from a national database or a trusted server, and the app will give out reminders on when to take as well as the important instructions like dosage.

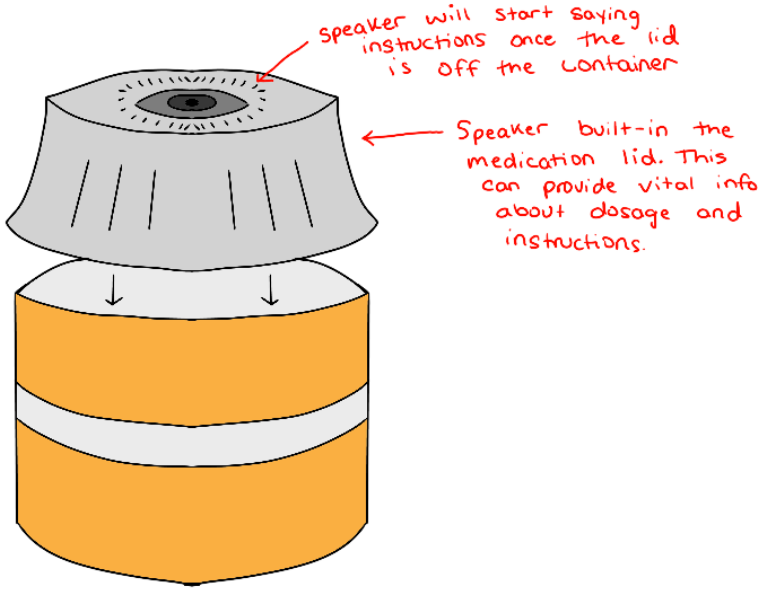


Medication Schedule App
Prescription Readability
Dhruv Shah Brainstorming #2

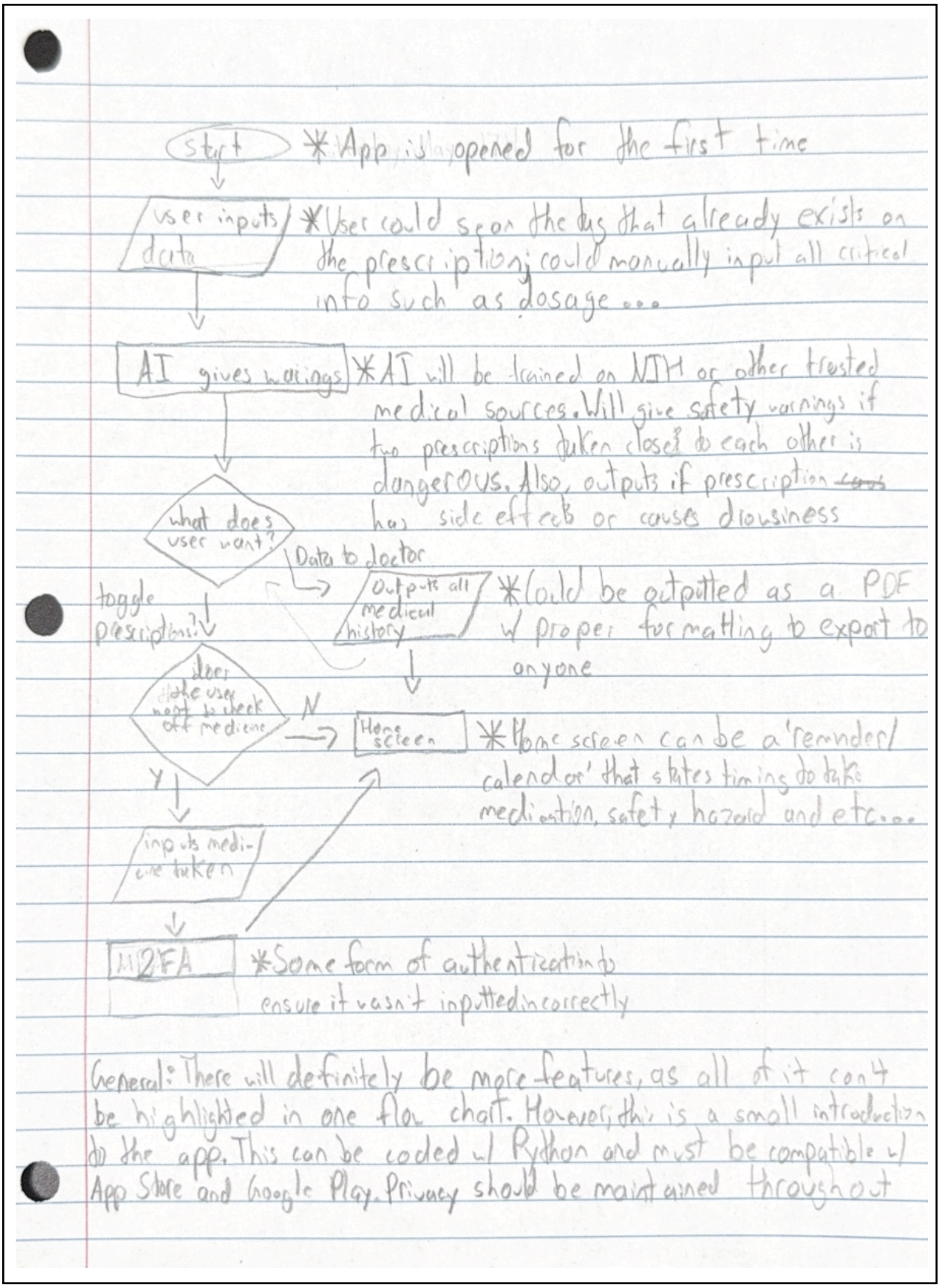


Microbit Scanner
Prescription Readability
Dhruv Shah Brainstorming #3


Pharmacy attaches QR code to label. Label is loaded with to text to speech algorithm of instructions and informations.
User can scan code with their phone to recieve information.
Using an HTML program, a website loaded with a text to speech algorithm button will present the info.
Generated QR Codes
Prescription Readability
Dhruv Shah Brainstorming #4

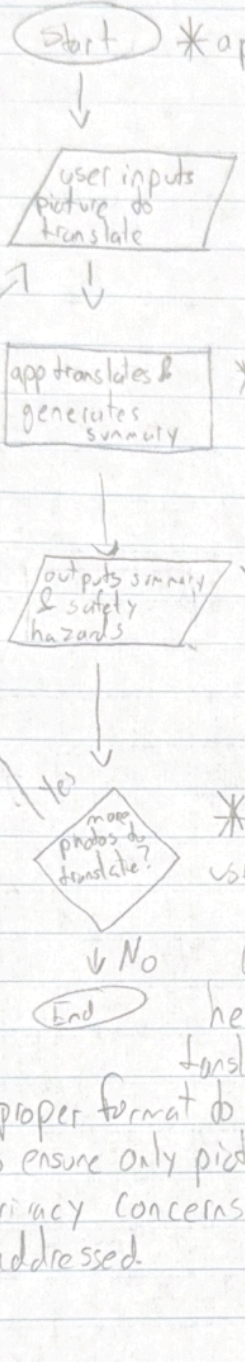


Built-in Speaker Lid
Prescription Readability
Dhruv Shah Brainstorming #5



General: There will definitely be more features, as all of it can't be highlighted in one flow chart. However, this is a small introduction to the app. This can be coded w/ Python and must be compatible w/ App Store and Google Play. Privacy should be maintained throughout

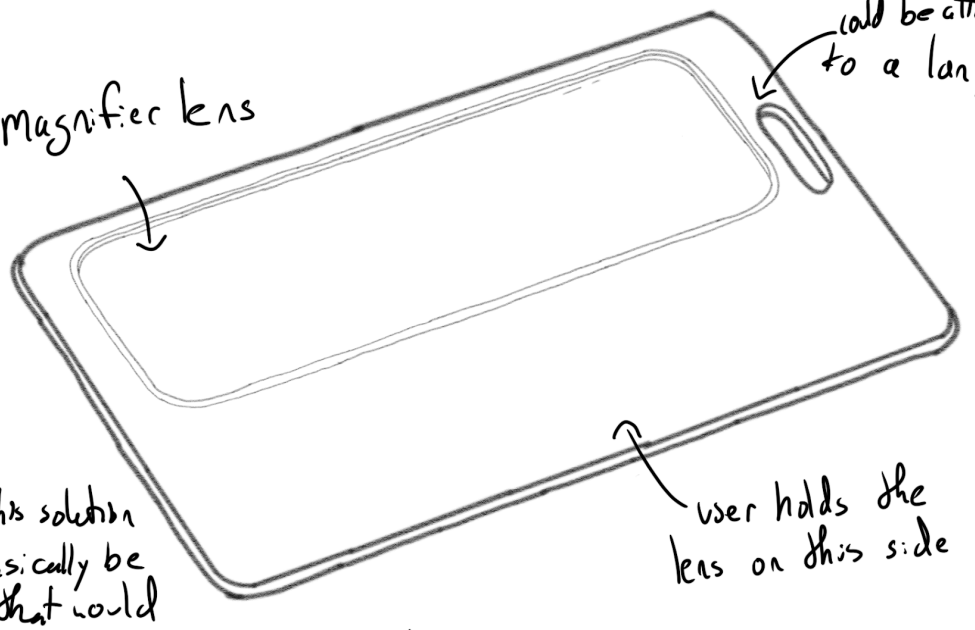
Krish Patel Medication Prescription Readability Sketch #2



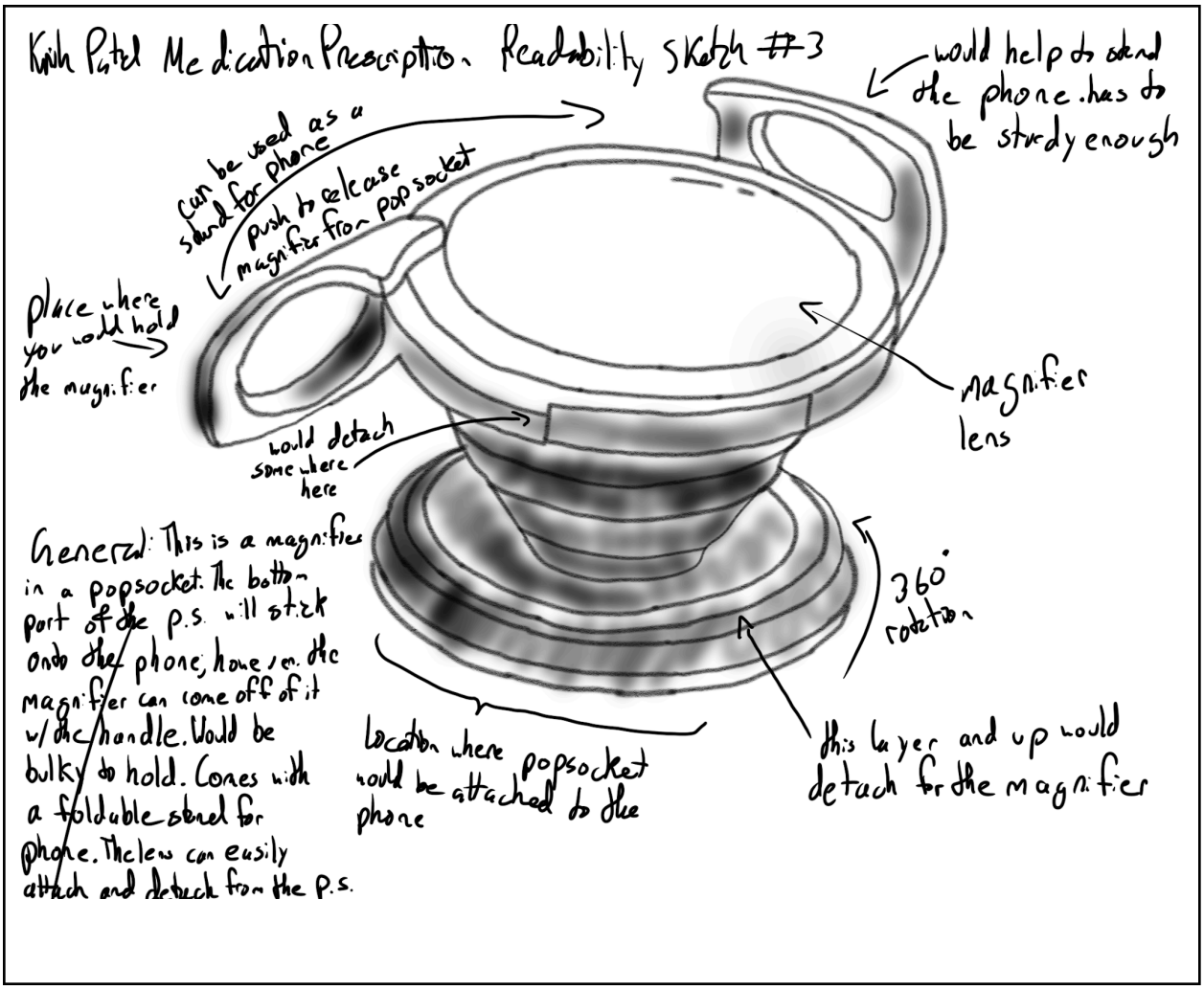
* app is opened
* there will be some form of guide that will teach the users on how to properly take photos
* the AI will be trained extensively on different images. will be similar to Google Translate. Some form of disclaimer will be given to highlight that there may be mistakes
* it will output the full text translated in whatever language the user selects. it will also output a quick summary of the critical info, such as expiry date and safety hazards.
* there will be some form of home screen where the user will be able do upload more photos

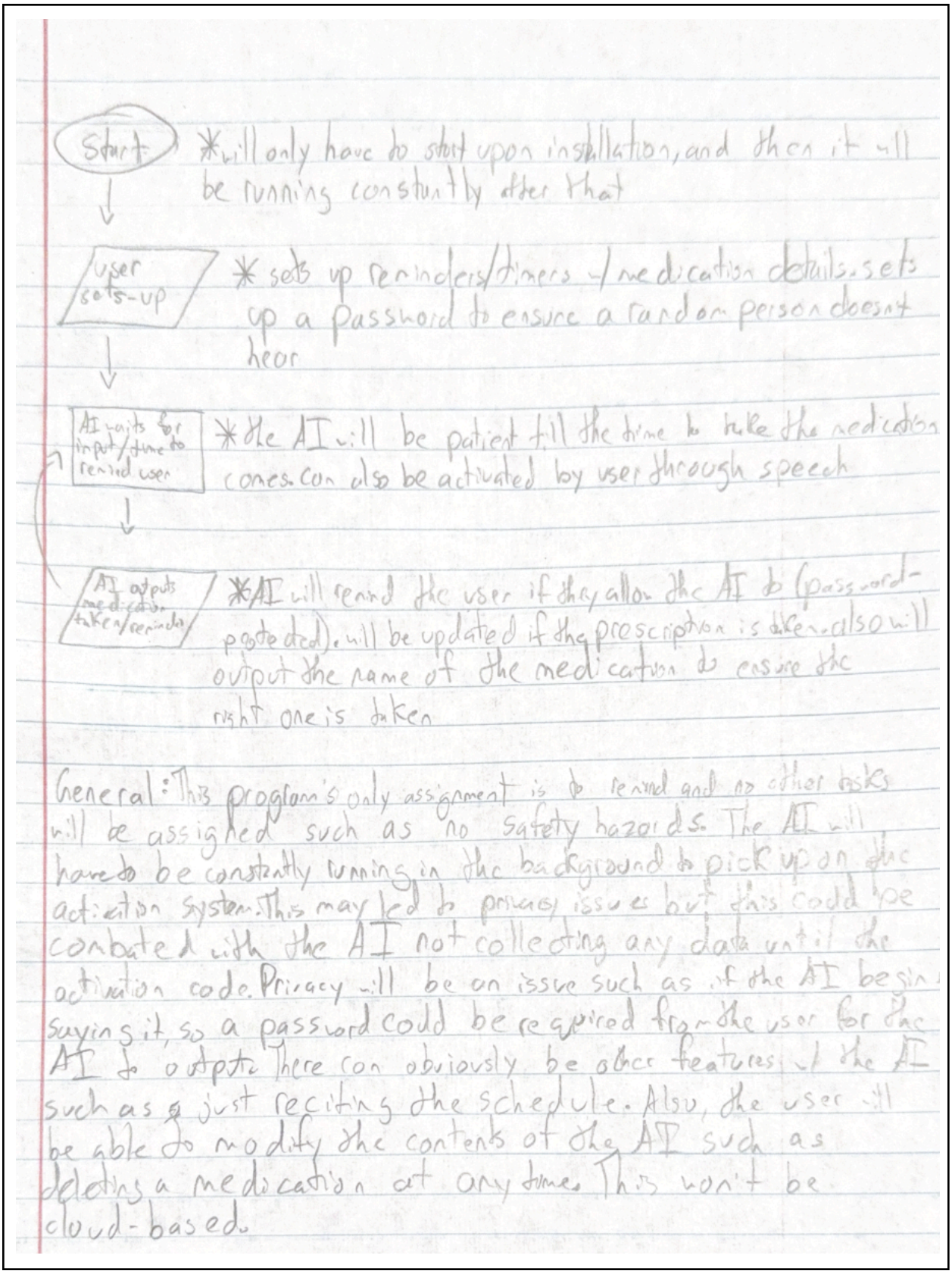
General: This AI will need to be really trained very heavily and will have to be validated on all translations to ensure mistakes are minimal. Would need a proper format to upload photos to ensure results are repeatable. Also to ensure only pictures of medications are uploaded. There will be privacy concerns of the images uploaded, so this will have to be addressed.

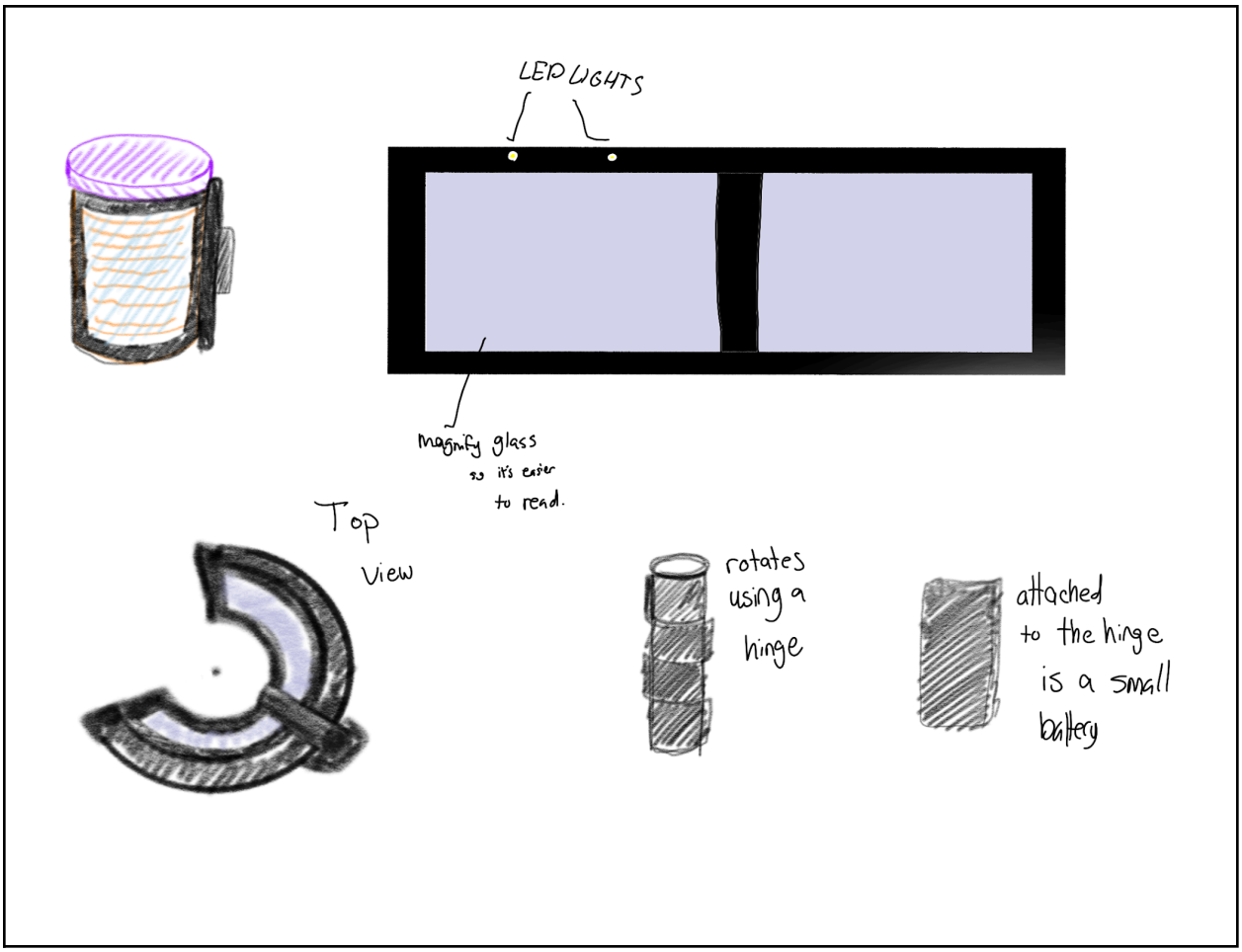
Krish Patel Medication Prescription Readability Sketch #4

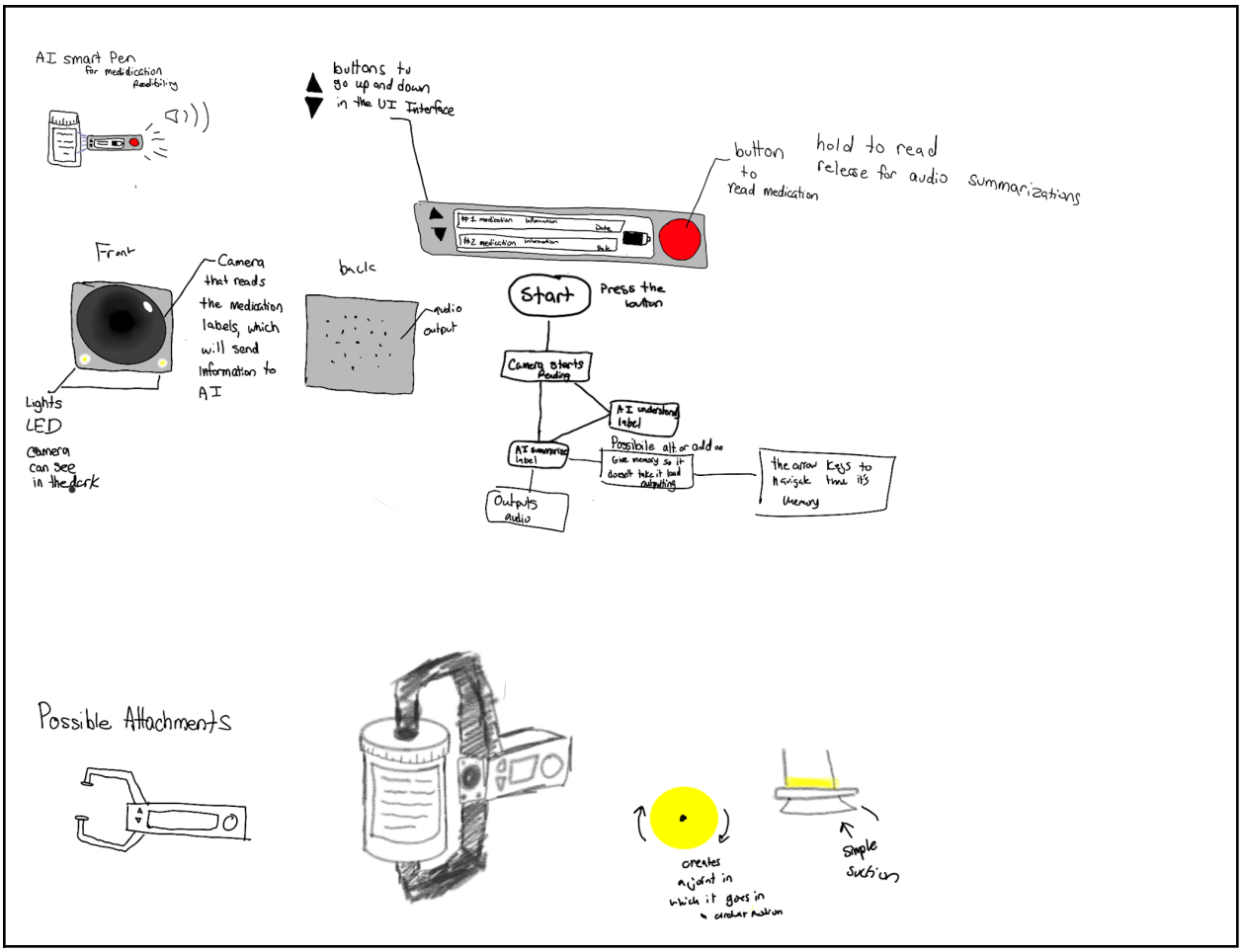


General: This solution would basically be a lens that would be thin enough to place in a wallet or a lanyard.
It would be pretty fragile so some sort of coating would be required around the lens.
The case of this solution is the one that has to be most worked on


Krish Patel Medication Prescription Readability Sketch #3
would help to extend the phone. has to be sturdy enough
can be used as a stand for phone
push to release magnifier from popsocket
place where you would hold the magnifier
magnifier lens
would detach somewhere here
General: This is a magnifier in a popsocket. The bottom part of the P.S. will stick onto the phone, however the magnifier can come off of it w/ the handle. Would be bulky to hold. Comes with a foldable stand for phone. The lens can easily attach and detach from the P.S.
360° rotation
location where popsocket would be attached to the phone
this layer and up would detach for the magnifier


Start
* will only have to start upon installation, and then it will be running constantly after that
user sets-up
* sets up reminders/timers + medication details, sets up a password to ensure a random person doesn't hear
AI waits for input/time to remind user
* the AI will be patient till the time to take the medication comes. Can also be activated by user through speech
AI outputs medication taken/reminder
* AI will remind the user if they allow the AI to (password-protected). will be updated if the prescription is taken. also will output the name of the medication to ensure the right one is taken
General: This program's only assignment is to remind and no other tasks will be assigned such as no safety hazards. The AI will have to be constantly running in the background to pick up on the activation system. This may lead to privacy issues but this could be combated with the AI not collecting any data until the activation code. Privacy will be an issue such as if the AI begins saying it, so a password could be required from the user for the AI to output. There can obviously be other features of the AI such as just reciting the schedule. Also, the user will be able to modify the contents of the AI such as deleting a medication at any time. This won't be cloud-based.

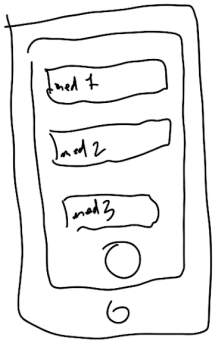

LED LIGHTS
magnify glass so it's easier to read.
Top View
rotates using a hinge
attached to the hinge is a small battery

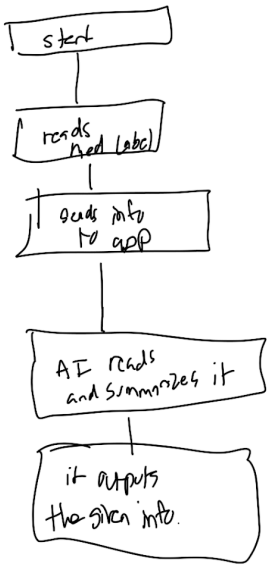

AI smart Pen
for medidiccation
Reading
buttons to
go up and down
in the UI Interface
button
to
read medication
hold to read
release for audio summarizations
Front
Camera
that reads
the medication
labels, which
will send
information to
AI
back
audio
output
Start
Press the
button
Camera Starts
Reading
AI understand
label
AI summarize
label
Possible alt or add on
Give memory so it
doesn't take it load
outputting
the arrow Keys to
Navigate time its
Memory
Outputs
audio
Lights
LED
Camera
can see
in the dark
Possible Attachments
creates
a joint in
which it goes in
a circular motion
simple
suction

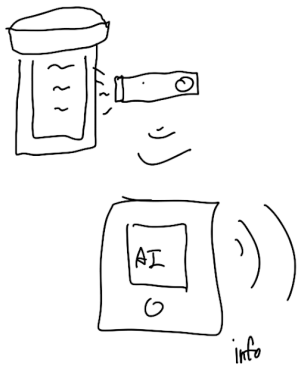
Continued from AI Pen

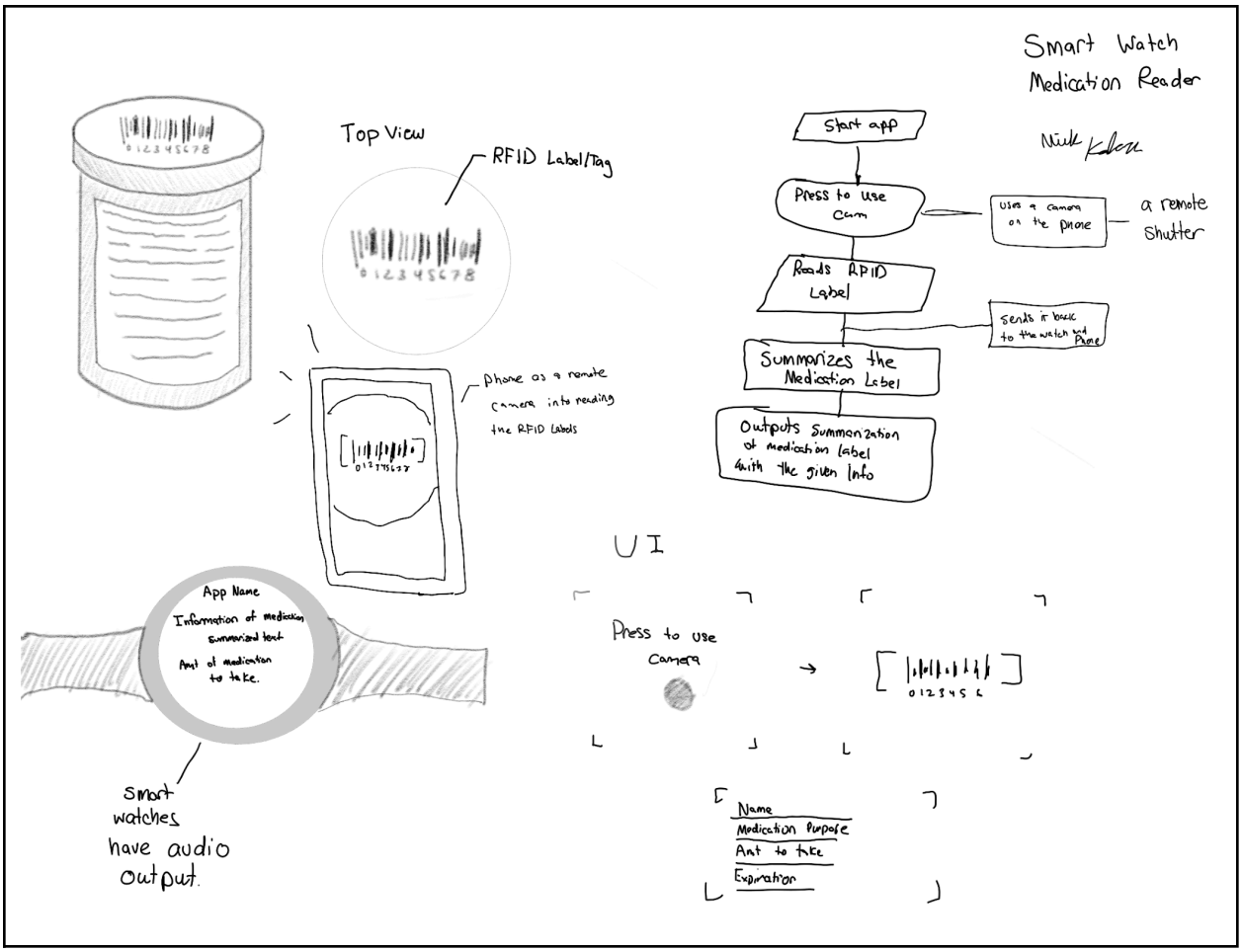
Same Pen

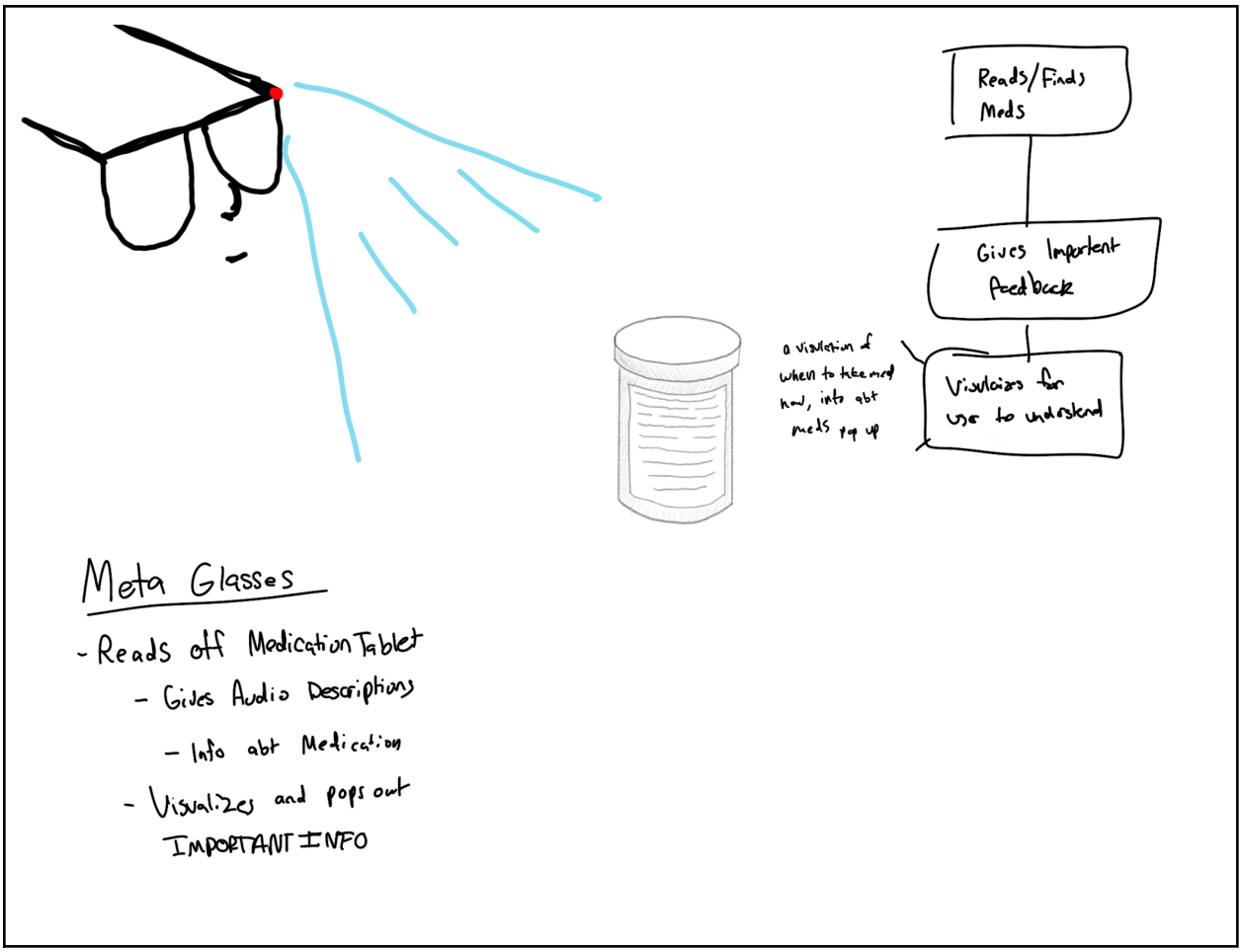
instead of AI inside of pen it only reads info and sends it to an digital app.

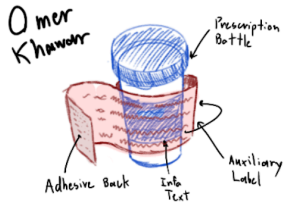




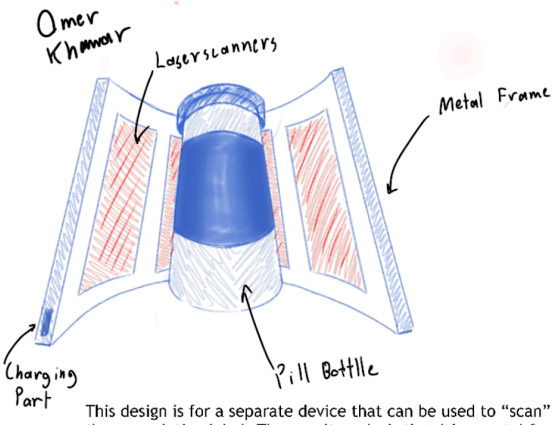



Smart Watch Medication Reader
Nick Katara
Top View
RFID Label/Tag
Phone as a remote camera into reading the RFID Labels
Start app
Press to use Cam
Uses a camera on the phone
a remote shutter
Reads RFID Label
Sends it back to the watch and Phone
Summarizes the Medication Label
Outputs Summarization of medication label with the given info
App Name
Information of medication summarized text
Amt of medication to take.
Smart watches have audio output.
UI
Press to use Camera
Name
Medication Purpose
Amt to take
Expiration

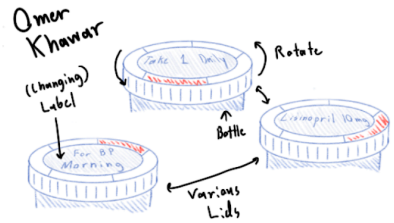

Reads/Finds Meds
Gives Important Feedback
Visualizes for user to understand
a visualization of when to take med how, info abt meds pop up
Meta Glasses
- Reads off Medication Tablet
- Gives Audio Descriptions
- Info abt Medication
- Visualizes and pops out IMPORTANT INFO



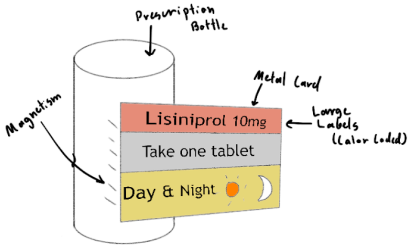
This design allows for an auxiliary prescription label. The way it works is by using a separately printed sheet/label that only contains the most valuable information for the user of the pills. Then, using the adhesive on the back of the paper it can be wrapped around the bottle to cover up the otherwise overwhelming text. This allows for larger fonts and reduced cognitive overload, making labels practical.



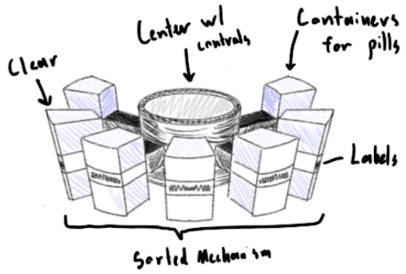
This design is for a separate device that can be used to "scan" the prescription label. The way it works is that it's a metal frame designed to encase the prescription bottle, and using laser scanning and AI it can identify the prescription without the need for existing RFID technology. It can take pull information from national databases to provide information to the user, either through an app or via audio.



This design uses rotating lid caps in order to display one category of information at a time. The user rotates the lid to switch between information on name, dosage, timing, etc. This solves the problem by reducing cognitive overload and allowing for larger fonts.



This design is for a metal card that is also magnetic. It can be attached to the prescription bottle so that it doesn't get lost, and on the card it contains very general information about the prescription that is most useful for the reader.



This design is for a pill organizer. The way it works is through one big control center that each capsule is attached to, which can then allow for further control of the medicines. It'll also allow for the user to read aloud which medicine is contained in which capsule out loud, along with easily modified labels.

Problem Statement

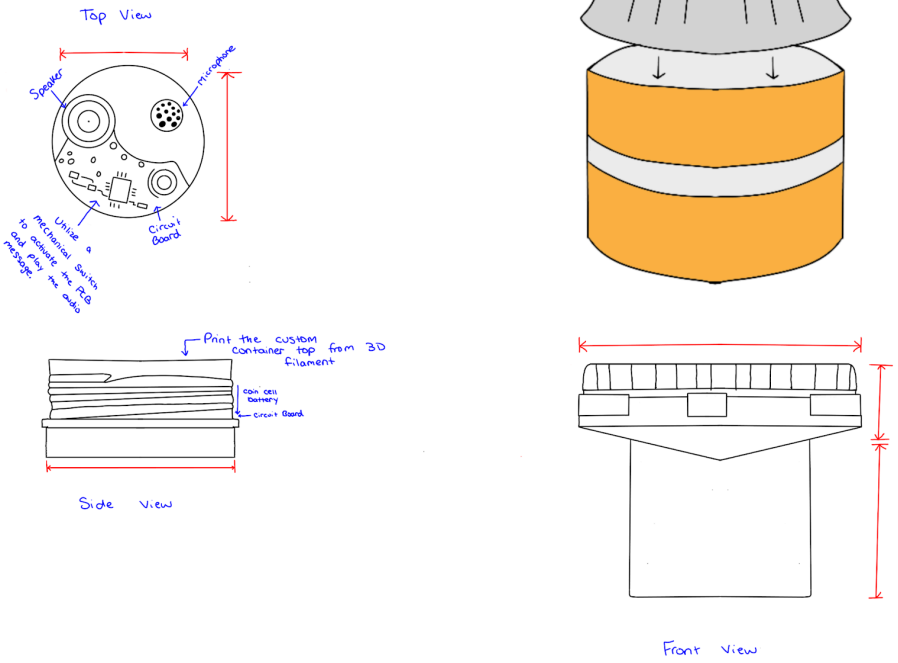
Older adults often struggle to read and understand prescription labels due to age-related vision decline, small font sizes, poor label design, and complex medical instructions. Studies show that between 76 and 81 percent of older adults do not meet the visual requirements to read labels. 18% of individuals already experience medication errors. These challenges are greater for individuals managing multiple prescriptions, especially when pills appear similar or labels are in unfamiliar languages. Current solutions, such as pillboxes, caregivers, and talking RFID labels, offer support but are often costly, inaccessible, or require technological familiarity. This issue primarily occurs at home, where older adults manage medications independently. Therefore, there is a clear need for an affordable and accessible solution that improves label clarity, readability, and usability to reduce errors and enhance medication safety.

Design Criteria

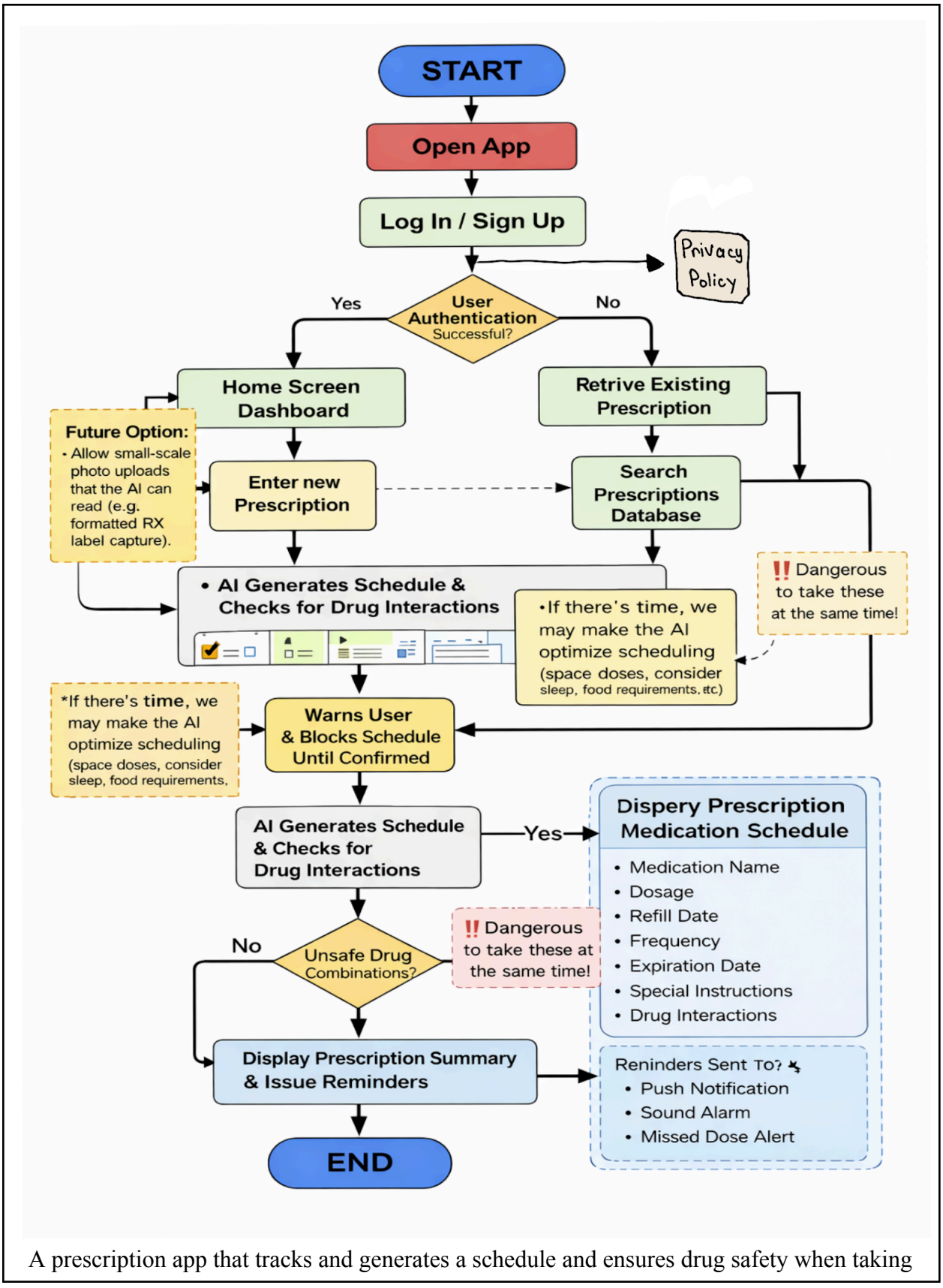
1. Safe & Accurate Use
2. High Readability

3. Ease of Use
4. Confidence - Building

Final Three Concept Ideas

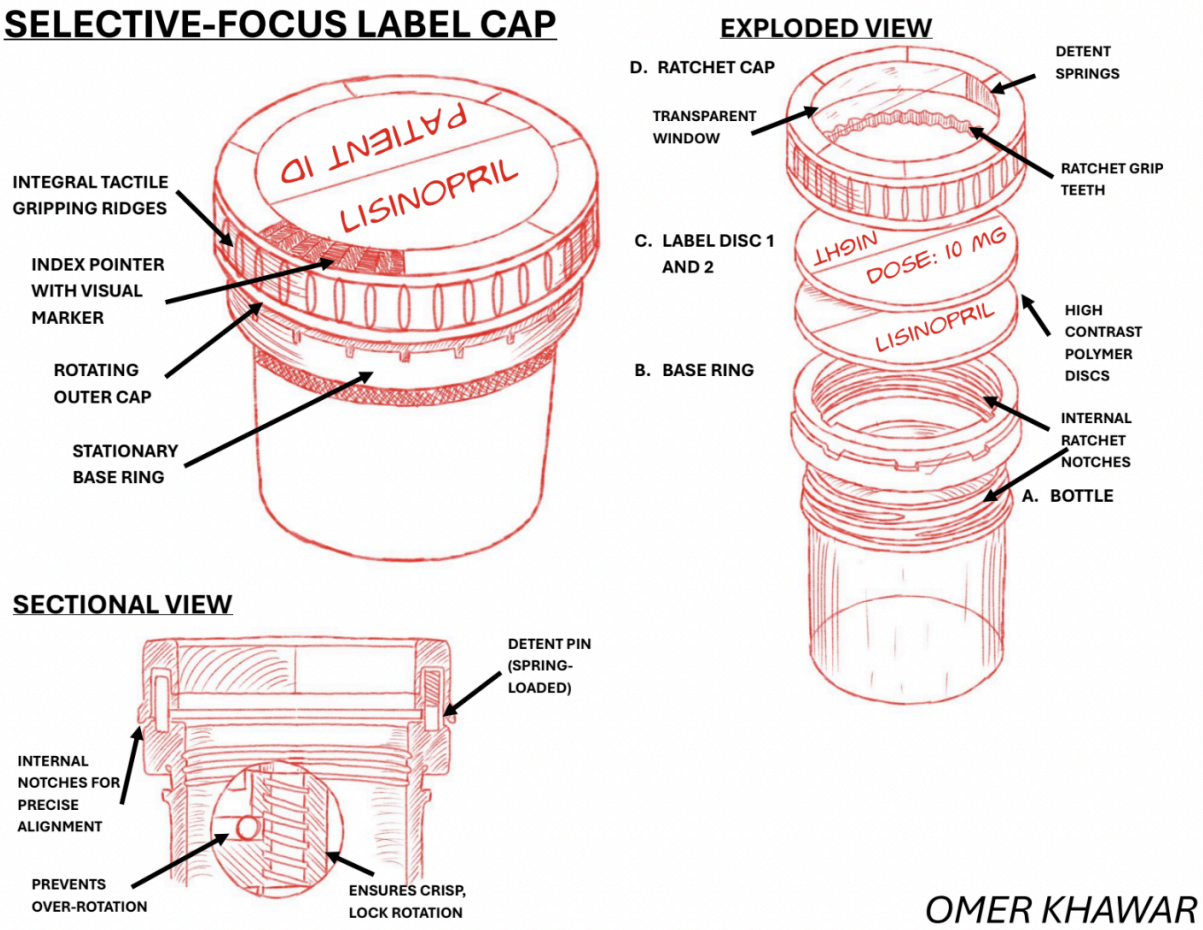


The design has a mechanical switch that activates the PCB to play the audio message. This message will be played from the speaker. The speaker and PCB will only be possible with a circuit board. There is a microphone attached; however, we are actively looking into how we can incorporate this and make it a functioning tool.



A prescription app that tracks and generates a schedule and ensures drug safety when taking

multiple prescriptions. The user will manually enter the prescription information. The app will remind the user of the prescription name, dosage, refill date, expiry date, and any special instructions. The app will also be able to set reminders to remind users to take medication. This will help users remember to take their medication if they have some form of memory loss



This product allows for a larger label to be displayed on the bottle cap of a prescription bottle. This label is very concise, containing only the most precise pieces of information, allowing for the largest font sizes possible. With a rotating cap design, users can then rotate the cap to then change what text is being displayed. In total, it allows for four different pieces of information to be displayed. This design doesn't replace the existing label, making sure information isn't lost.

Decision Matrix - 20 Solutions

Decision Matrix - 20 Solutions

The categories that we used to score the potential solutions are: Accessibility, Cost, Usability, Efficiency, and Development Time. We scored each category out of 5.

Final Three Concept Ideas

Final Three Concept Ideas Presentation

We presented this to our peers for constructive feedback.

Final Three Concept Ideas Survey

We will share this survey with other individuals so we can collect even further feedback.

Element D3 - Final Concept Selection Documentation

Element D3 - Final Concept Selection Documentation

Attached is the Element D3 documentation that contains: product description, operation, justification, consumer feedback, sketches/concepts, and more. This document will be readily available when individuals request the documentation of our final concept selection.

CAD

CAD Details

Attached are all of the CAD details such as the link to the Onshape, as well as the link to the Tinkercad circuit. The dimensions of the prescription bottle are also highlighted. The technical drawings of each part of the pill bottle will be added. Additionally, the circuit from the Tinkercad is visualized alongside the circuit for the circuit.

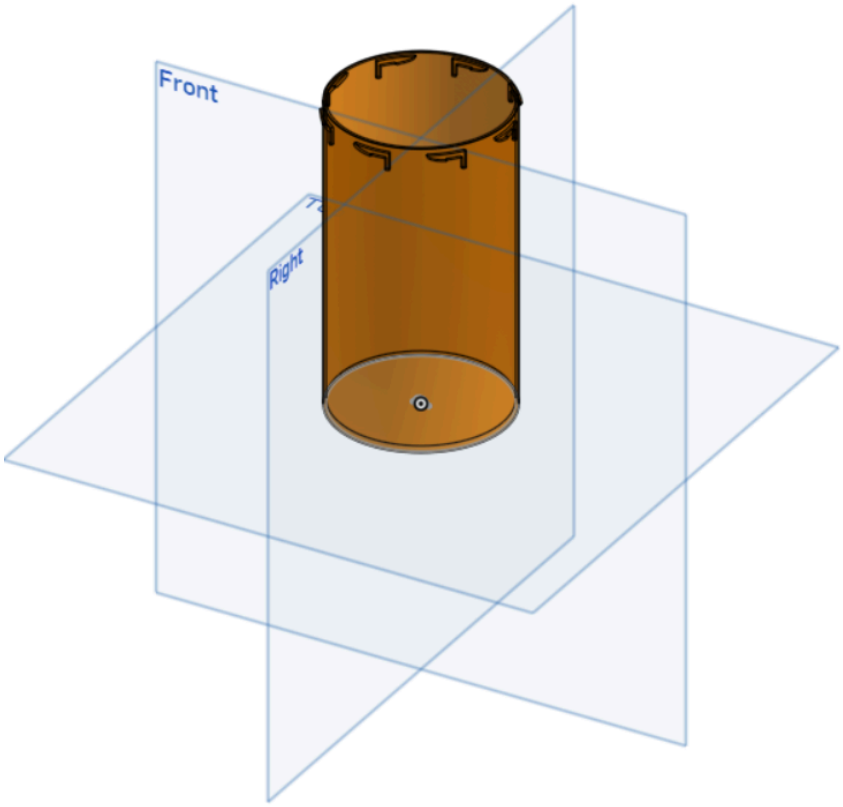
More pictures will be added down below.

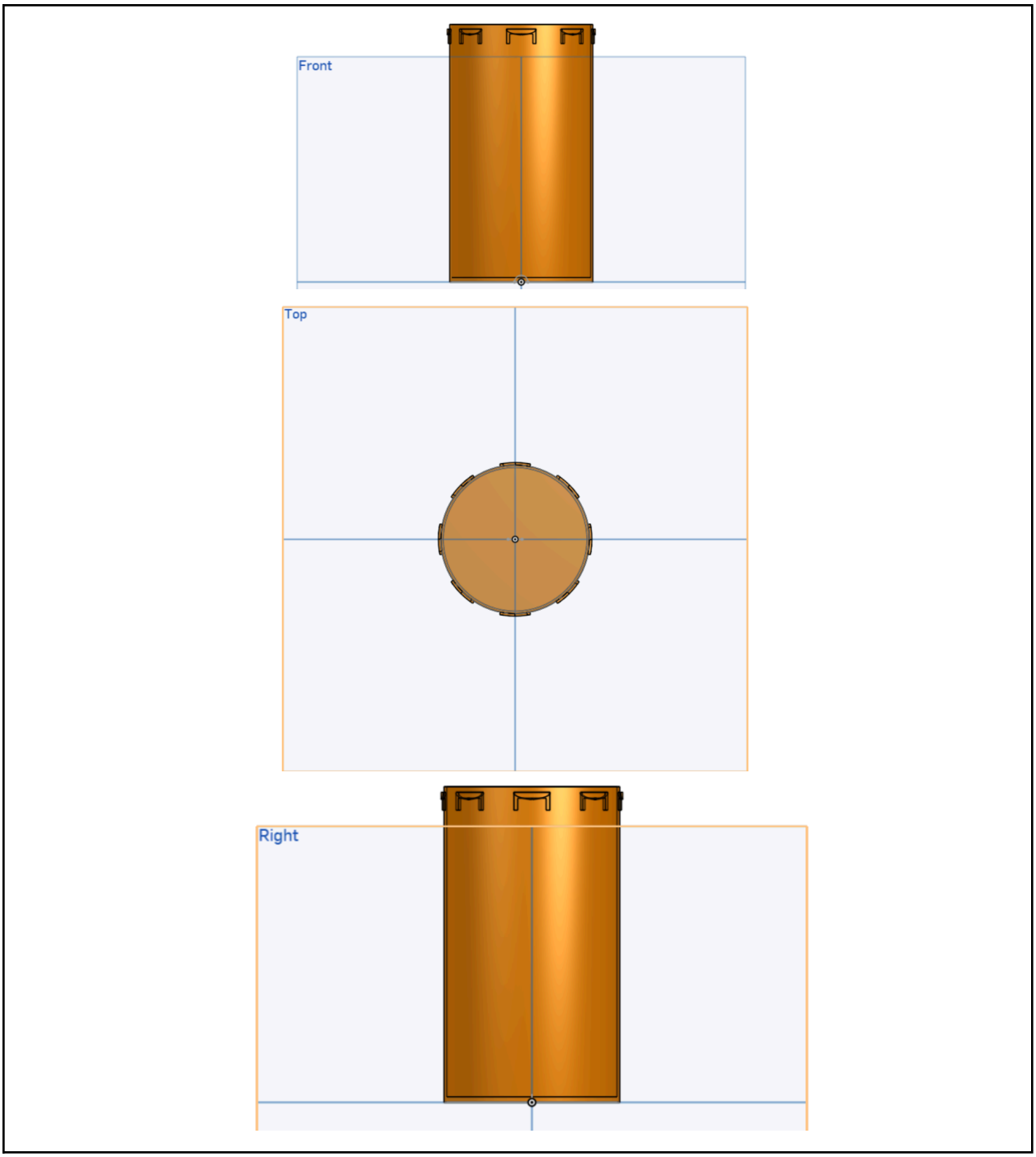
Materials List

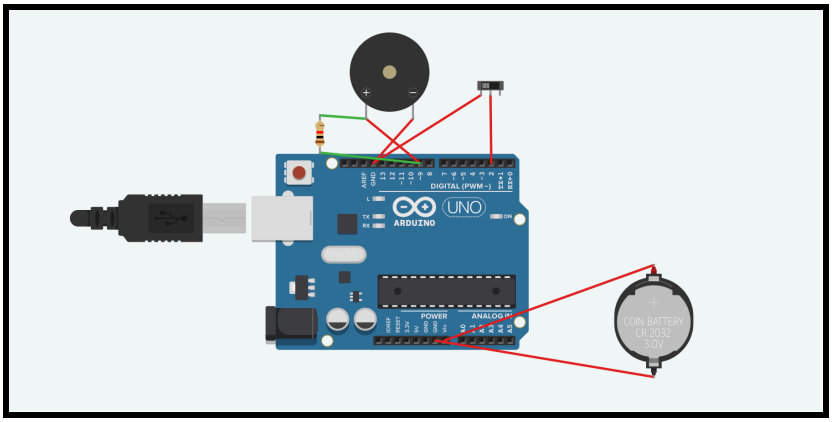
Materials List

Attached are all the expected materials that we will use to make the final product. More materials may be used, and likewise, less materials could be used. The purpose of each material is also highlighted.

Images of the CAD




Front
Top
Right

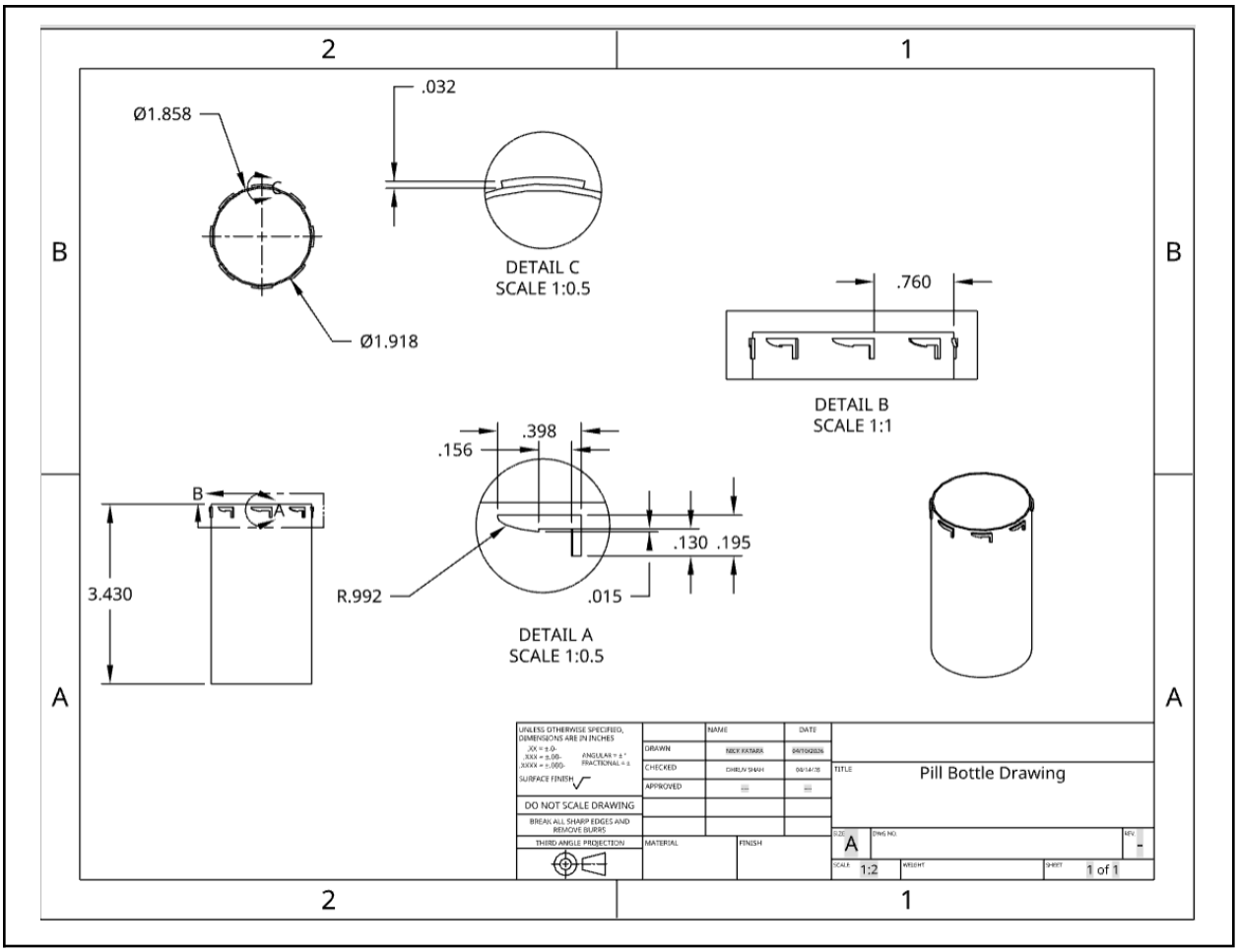


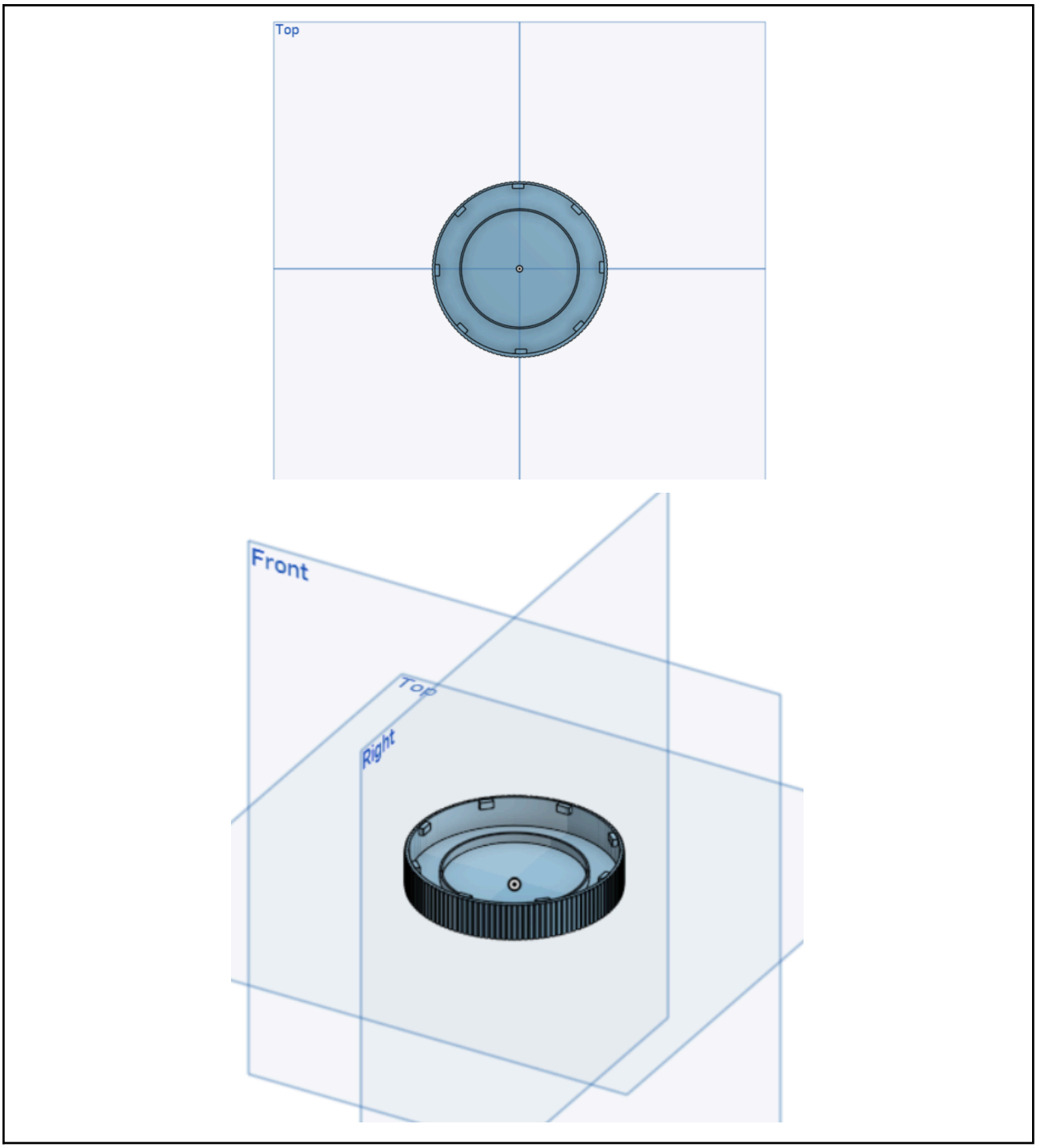
Sample Arduino Code:

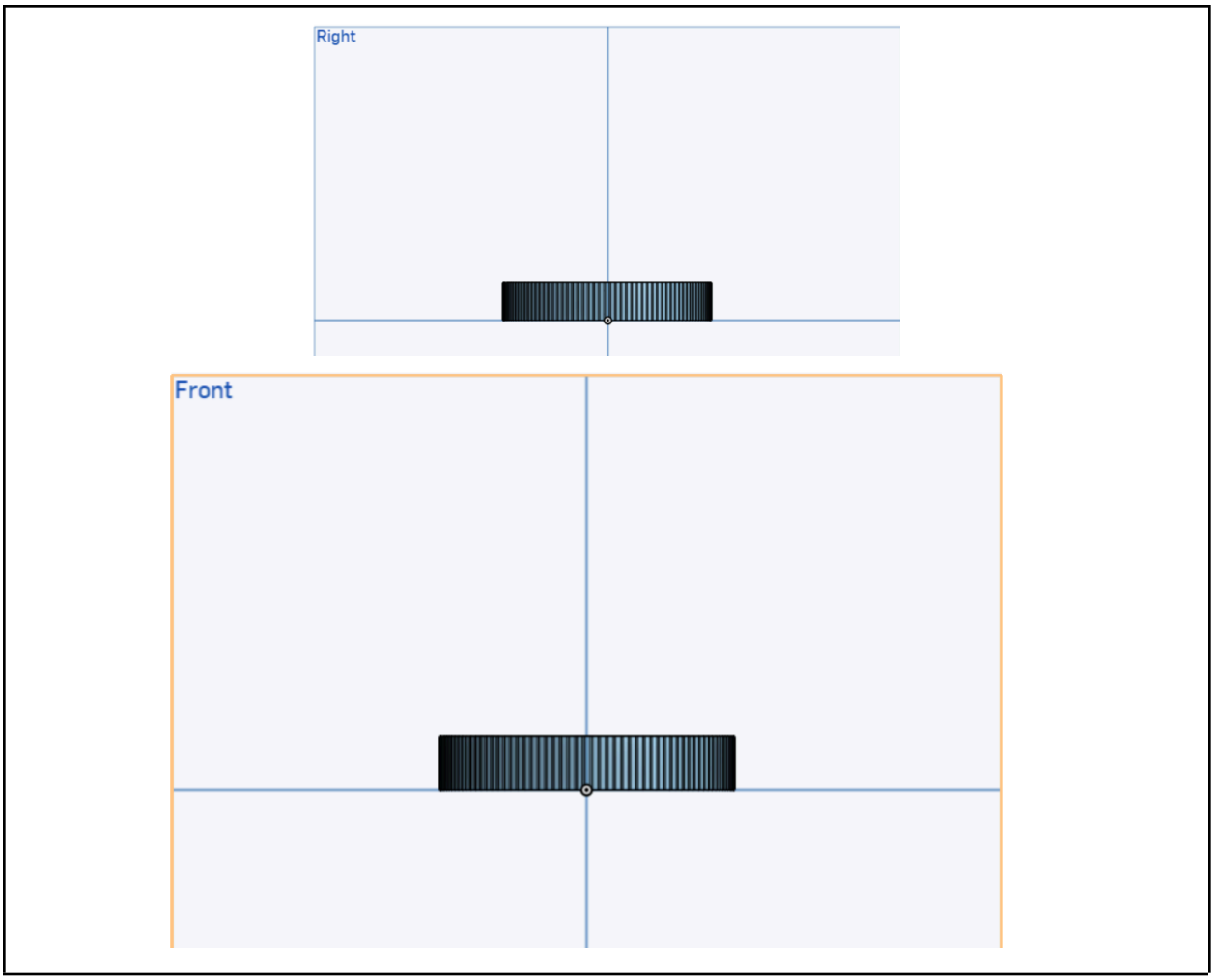
```
int lidPin = 2; // Pin connected to the switch

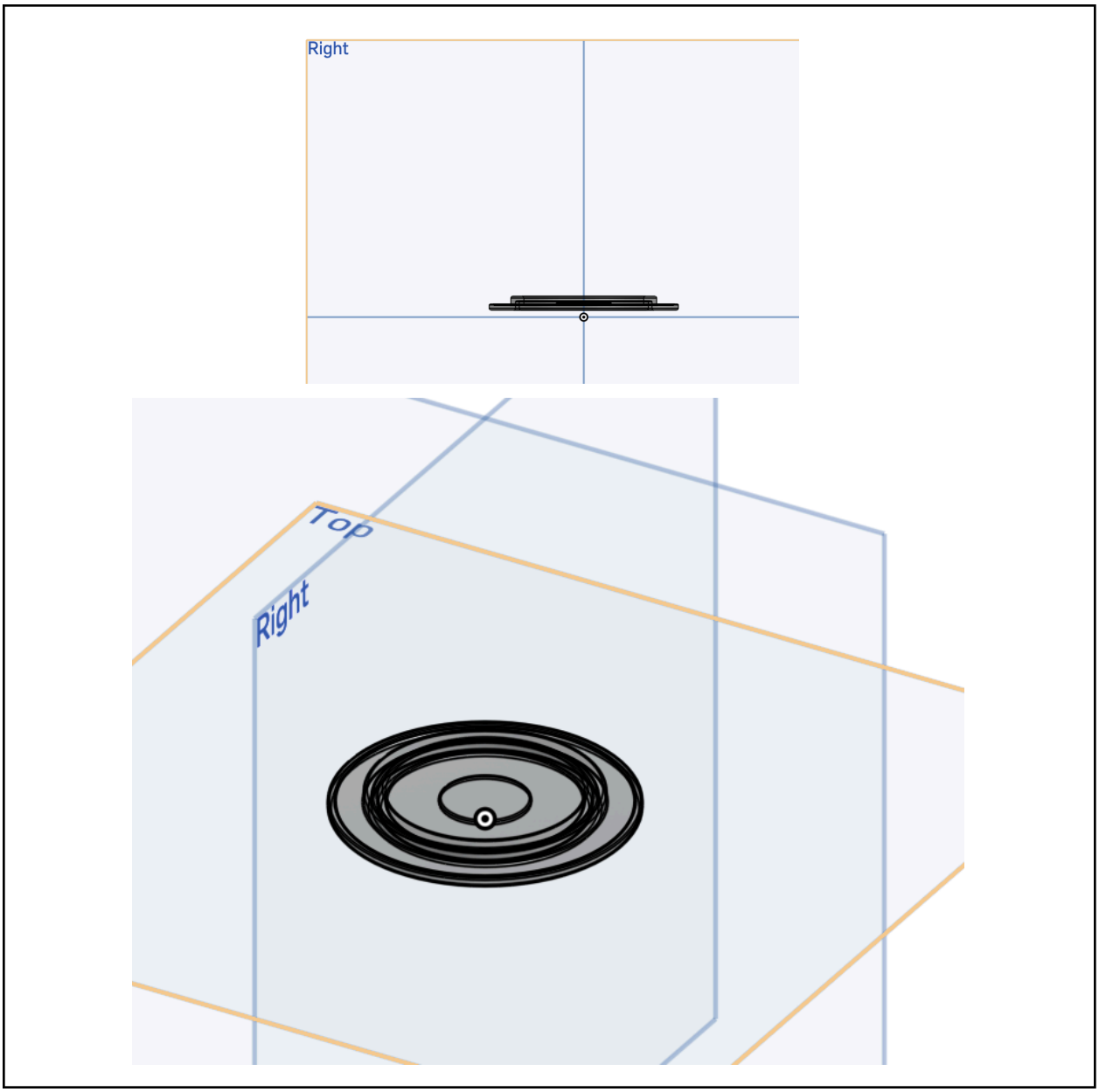
void setup() {
  pinMode(lidPin, INPUT_PULLUP); // Keeps pin HIGH until switch is flipped
  pinMode(9, OUTPUT);            // Pin for the Buzzer/Sound
}

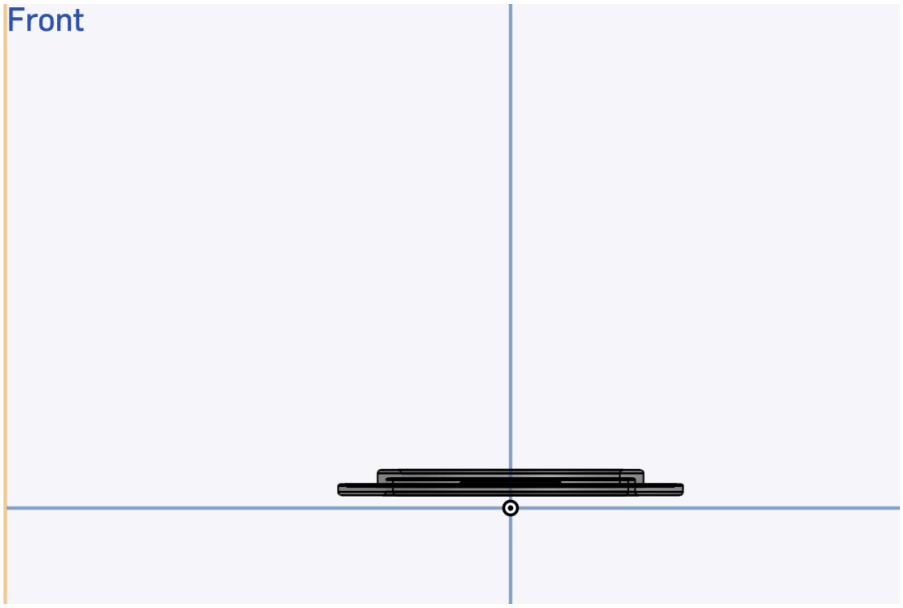
void loop() {
  // If the switch is moved (simulating lid removal)
  if (digitalRead(lidPin) == LOW) {
    // This "tone" represents your MP3 playing
    tone(9, 880, 1000); // Play a 440Hz tone for 1 second
    delay(5000);        // Wait 5 seconds so it doesn't loop constantly
  }
}
```

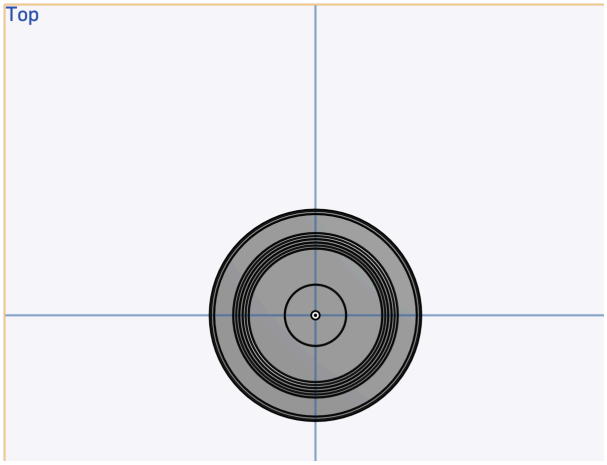

2
1
B
A
Ø1.858
C
Ø1.918
.032
DETAIL C
SCALE 1:0.5
.760
DETAIL B
SCALE 1:1
B
A
3.430
.156
.398
.130
.195
.015
R.992
DETAIL A
SCALE 1:0.5
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
NICK KATARA
CHECKED
DHRUV SHAH
APPROVED
MATERIAL
FINISH
TITLE
Pill Bottle Drawing
SIZE
A
DWG NO.
REV
SCALE
1:2
WEIGHT
SHEET
1 of 1

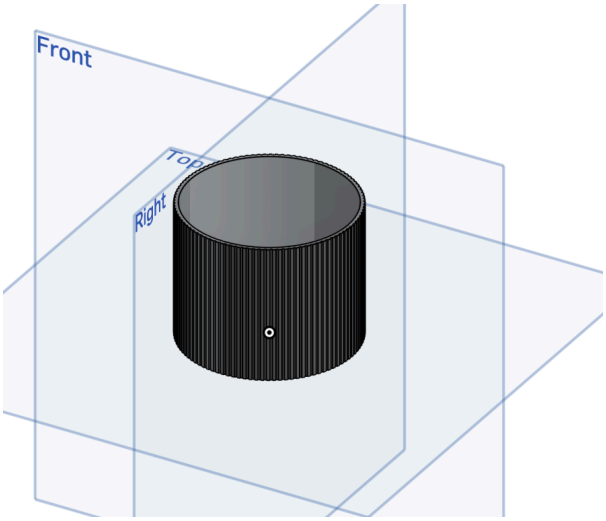

Top
Front
Top
Right

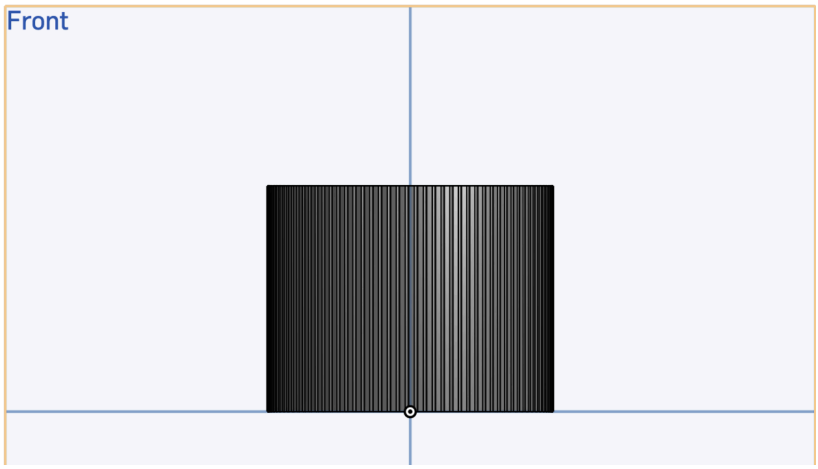

Right
Front

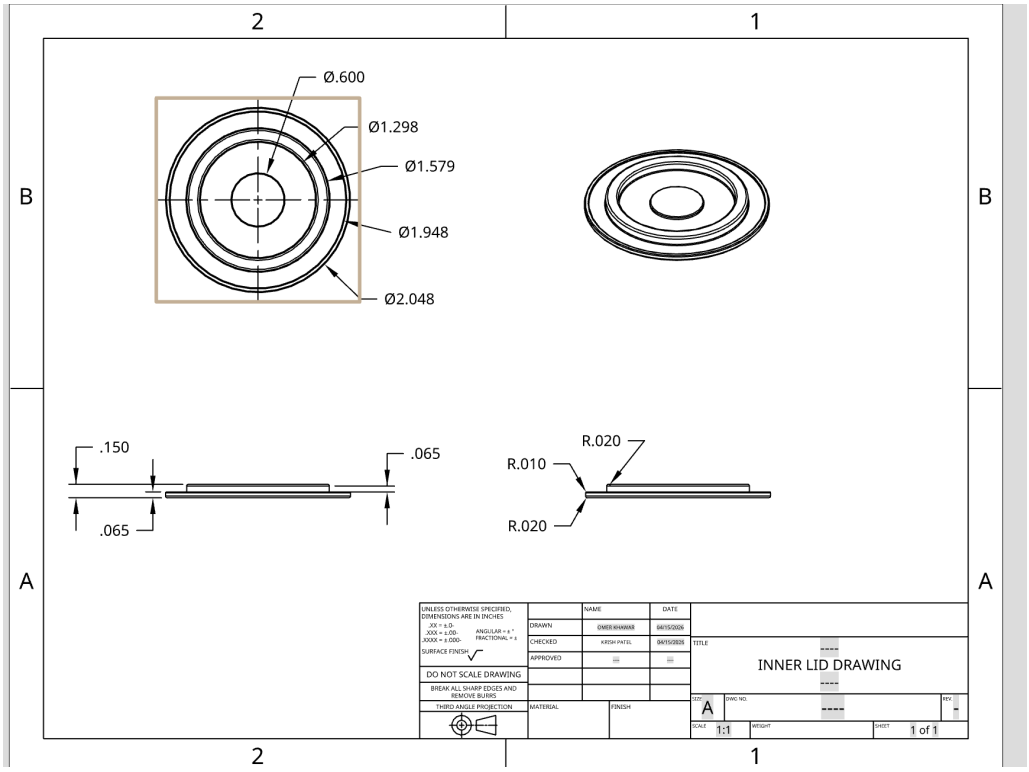

Right
Top
Right

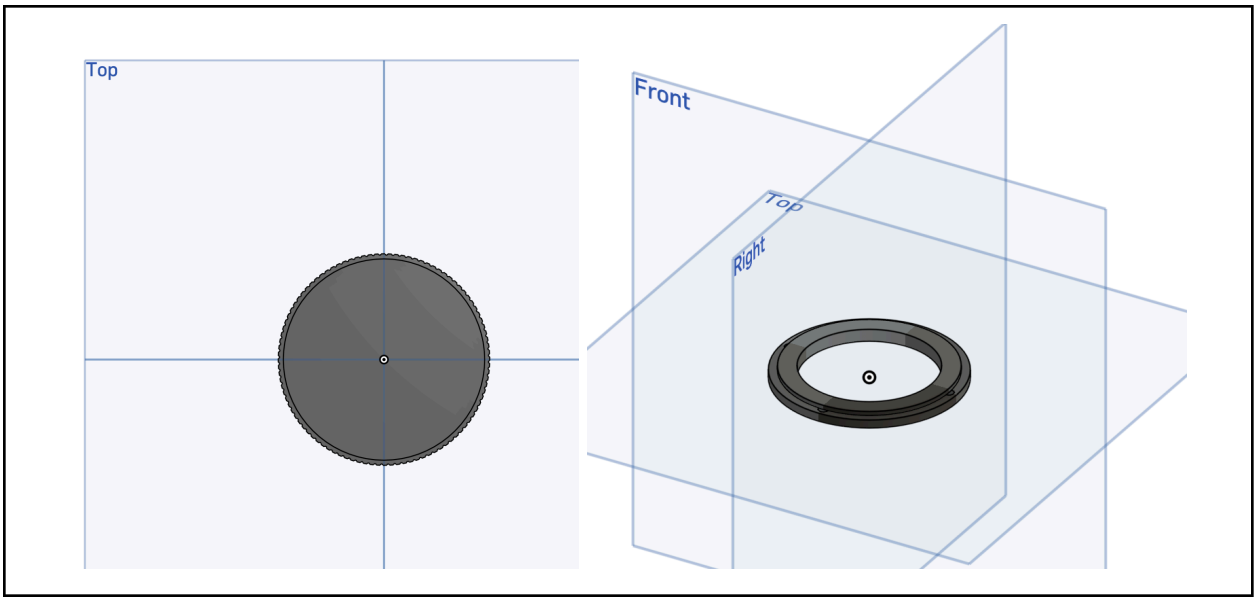

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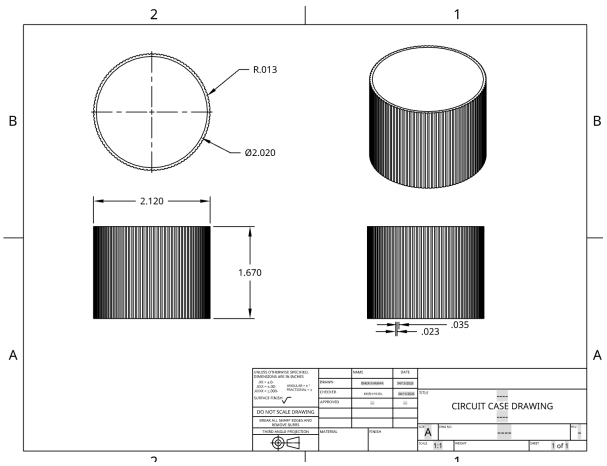

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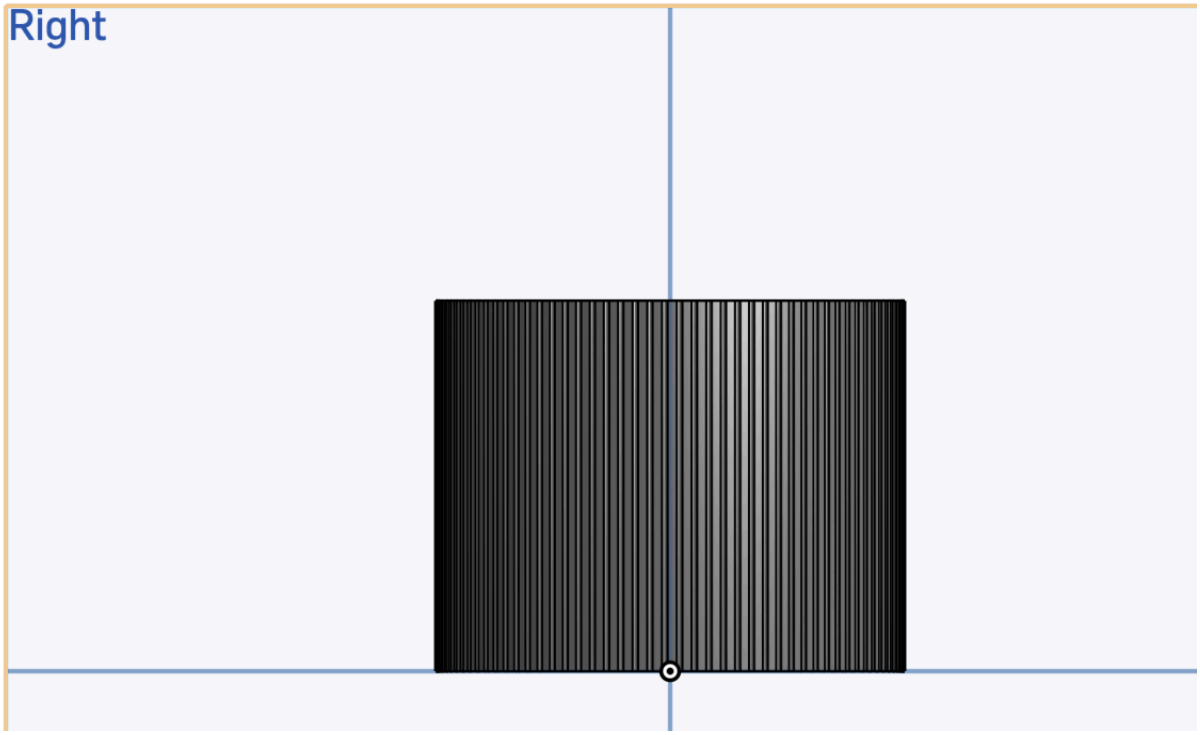

Front
Right


Front

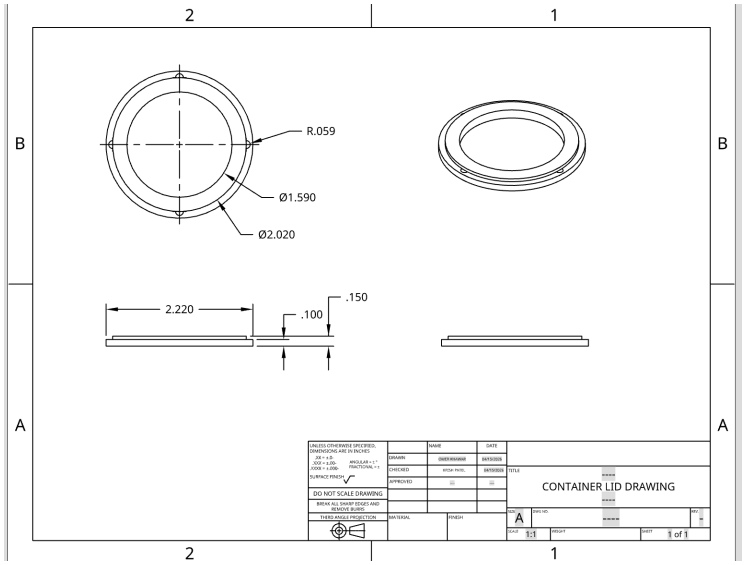

2
1
B
A
Ø.600
Ø1.298
Ø1.579
Ø1.948
Ø2.048
.150
.065
.065
R.020
R.010
R.020
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
OMER KHAWAR
04/15/2026
CHECKED
KRISH PATEL
04/15/2026
APPROVED
MATERIAL
FINISH
TITLE
INNER LID DRAWING
SIZE
A
DWG NO.
REV
SCALE
1:1
WEIGHT
SHEET
1 of 1

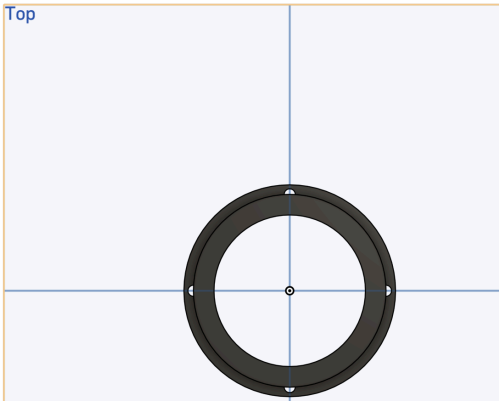

Top
Front
Top
Right


2
1
B
B
A
A
R.013
Ø2.020
2.120
1.670
.035
.023
CIRCUIT CASE DRAWING
DO NOT SCALE DRAWING

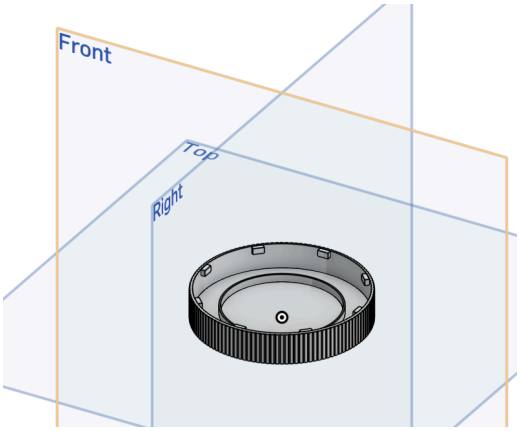

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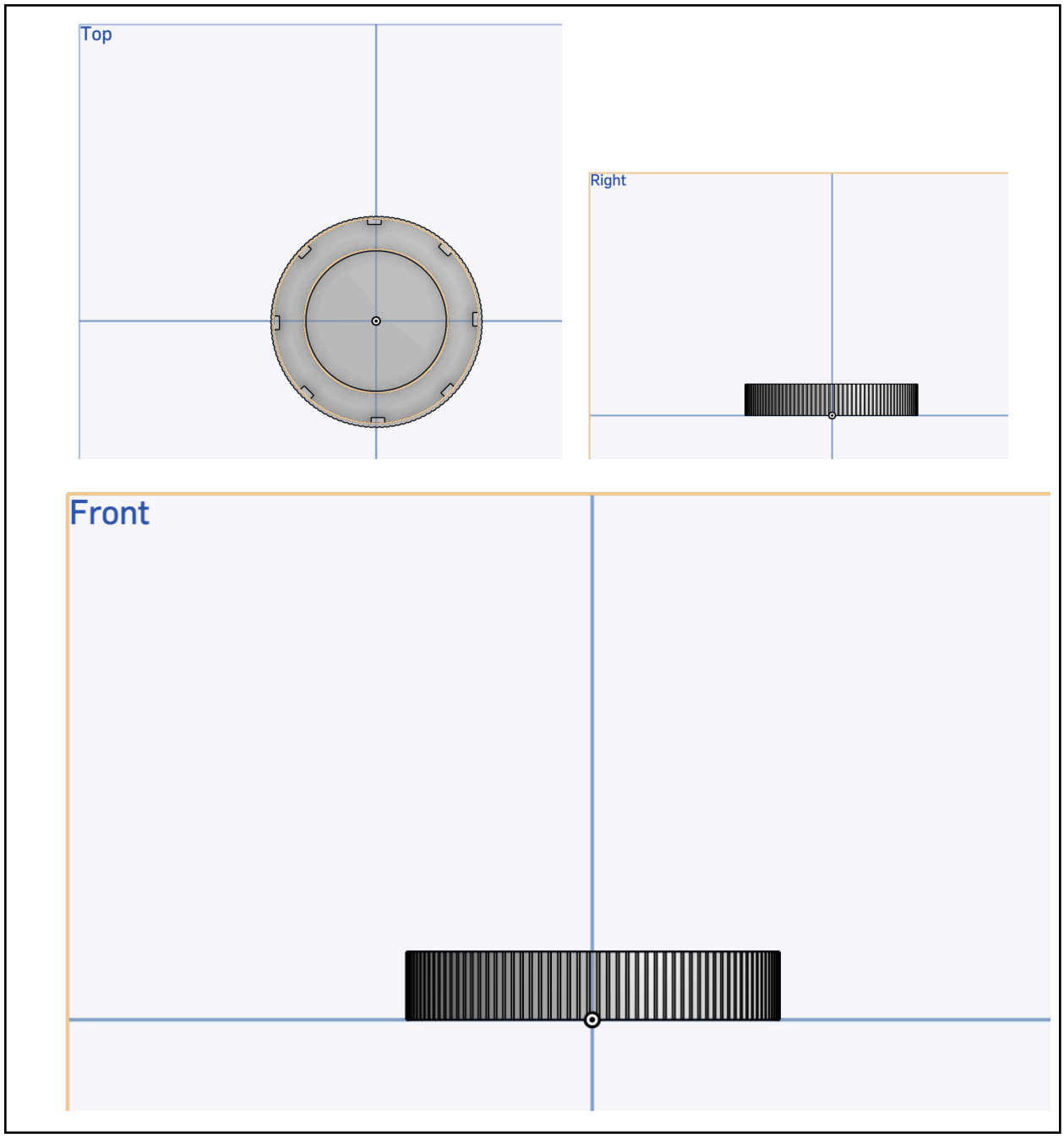

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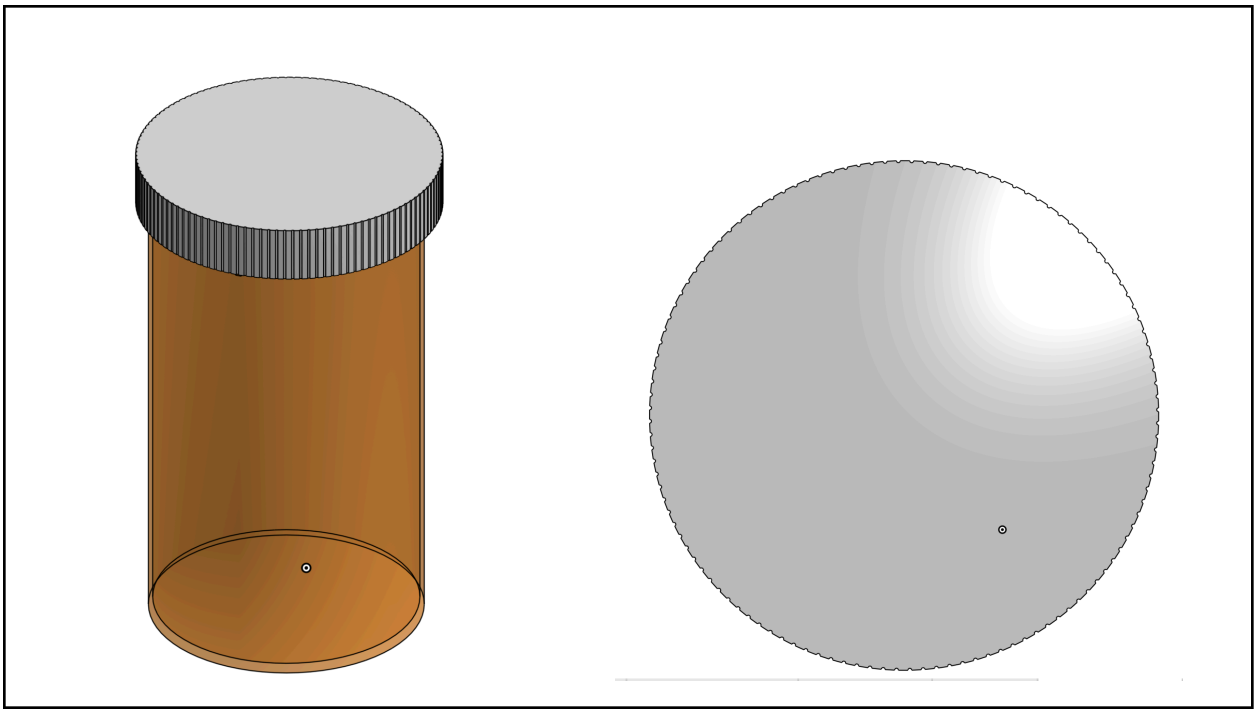

2
1
B
A
R.059
Ø1.590
Ø2.020
2.220
.100
.150
CONTAINER LID DRAWING
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION

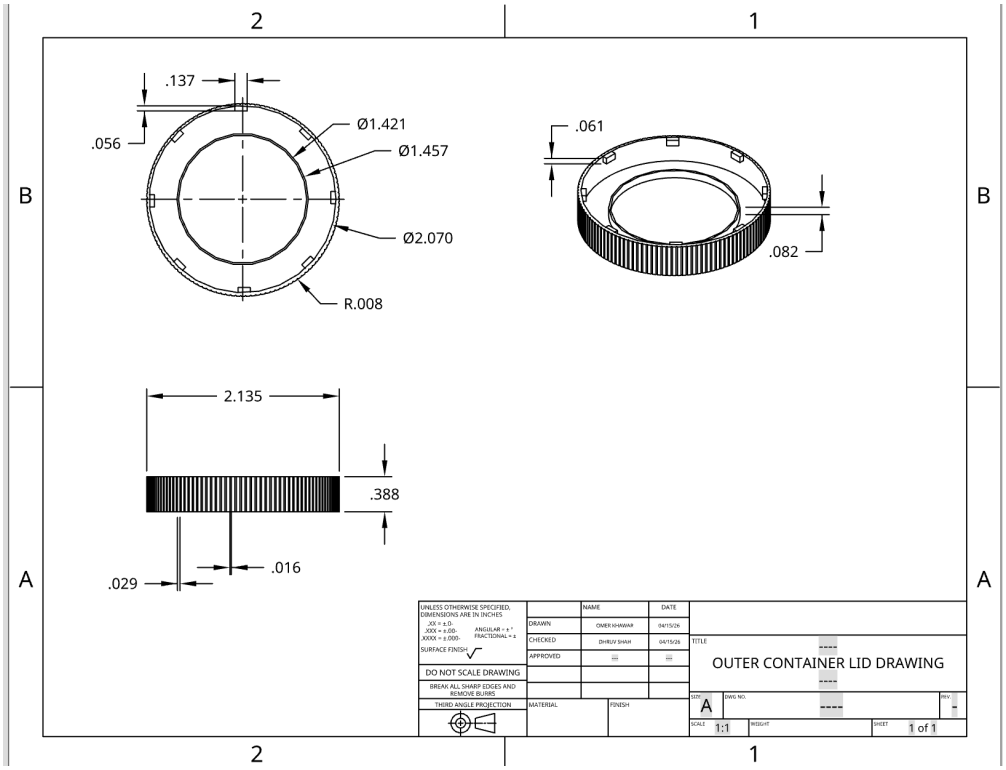

Top

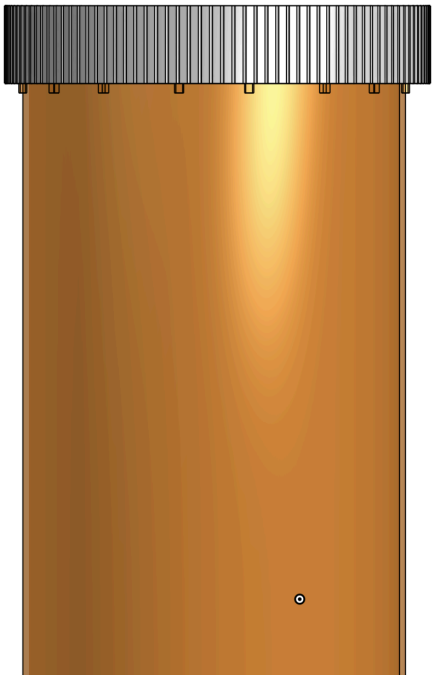

Front

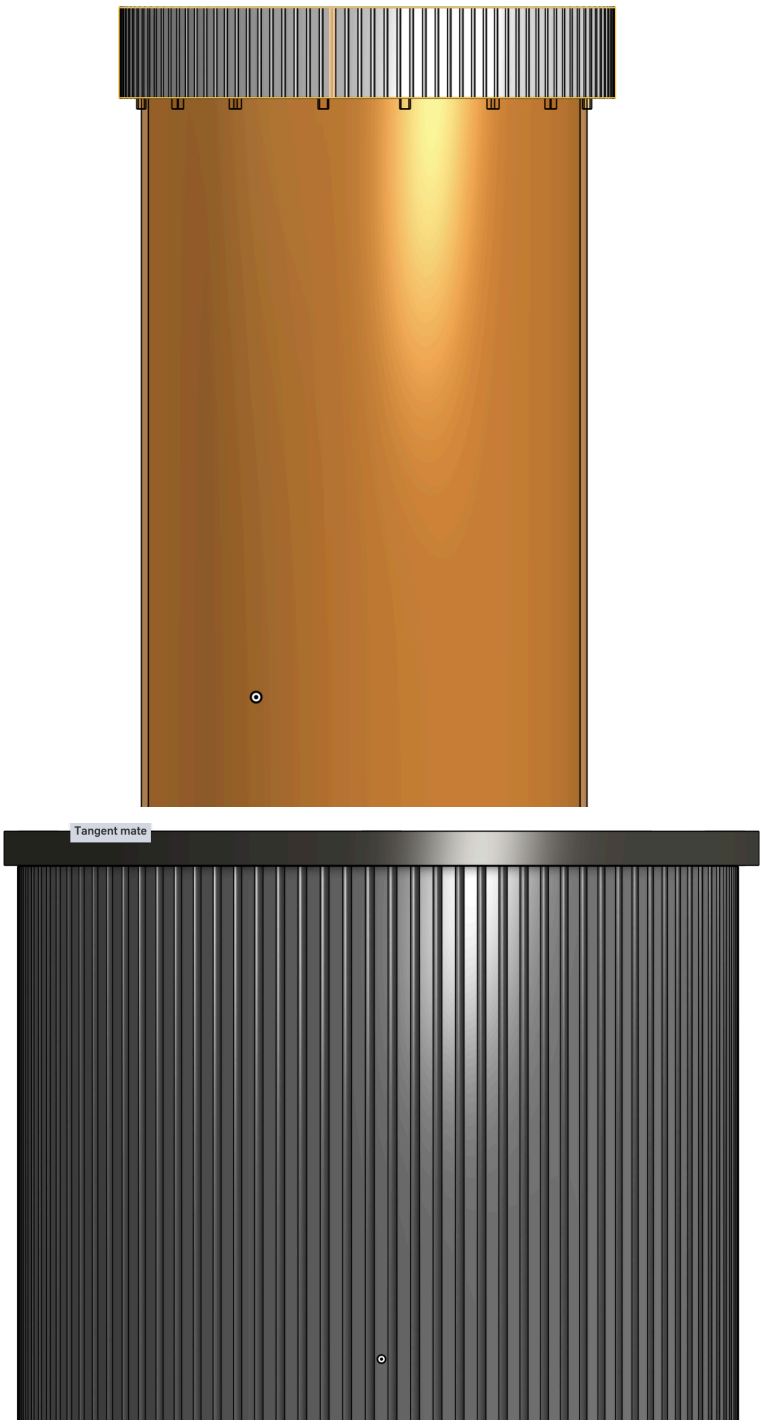

Front
Top
Right

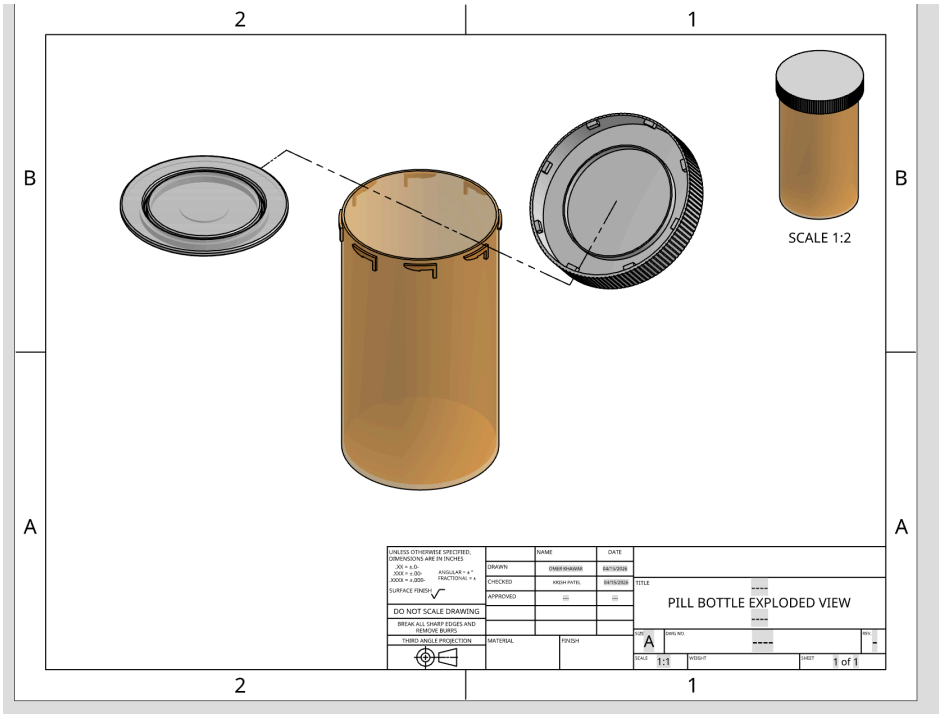

Top
Right
Front



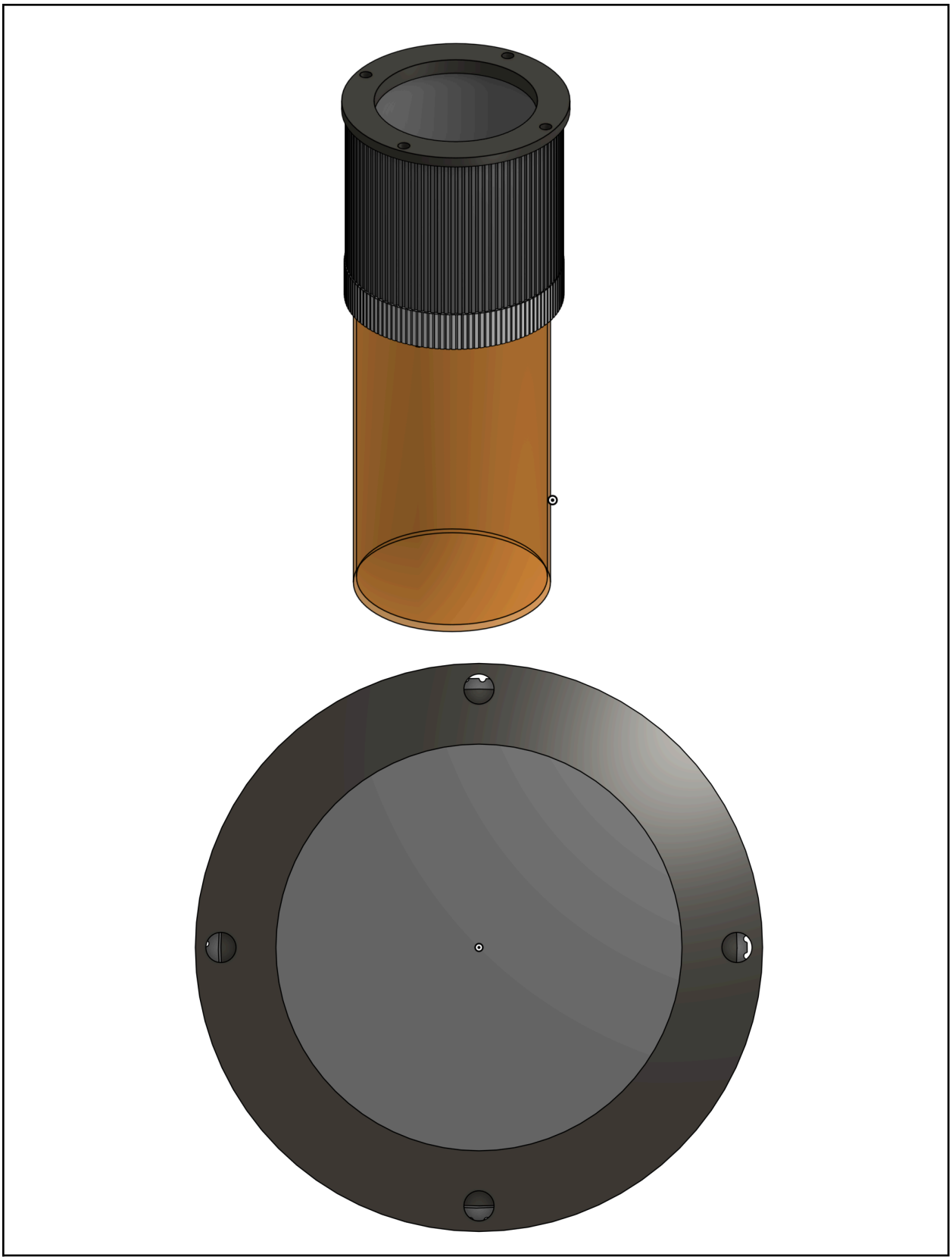

2
1
B
A
.137
.056
Ø1.421
Ø1.457
Ø2.070
R.008
.061
.082
2.135
.388
.029
.016
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX ± .05
.XXX ± .005
.XXXX ± .0005
ANGULAR ± °
FRACTIONAL ± ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
OMER KHAWAR
04/15/26
CHECKED
DHRUV SHAH
04/15/26
APPROVED
MATERIAL
FINISH
TITLE
OUTER CONTAINER LID DRAWING
SIZE
A
DWG NO.
REV
SCALE
1:1
WEIGHT
SHEET
1 of 1

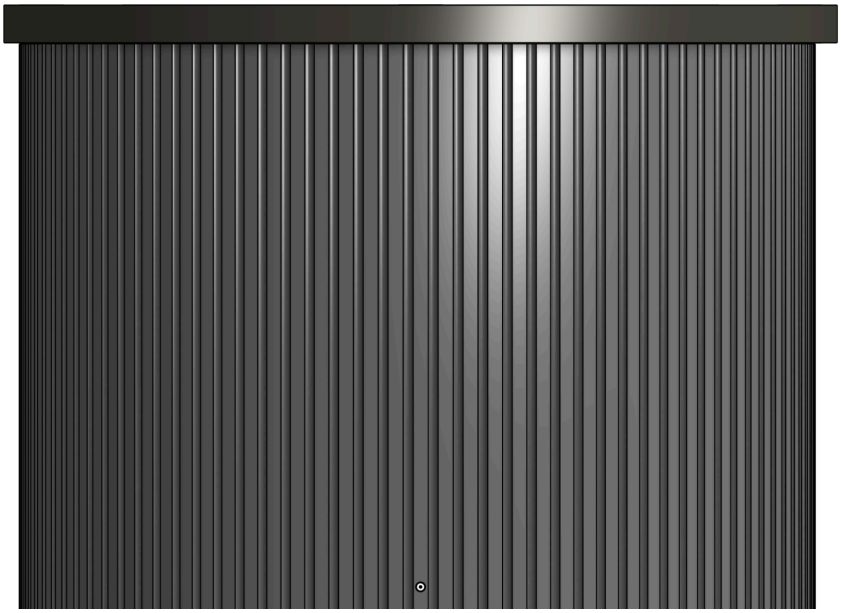


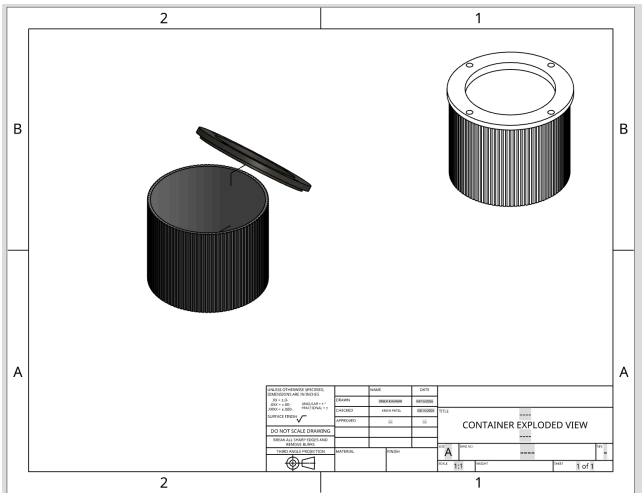

Tangent mate

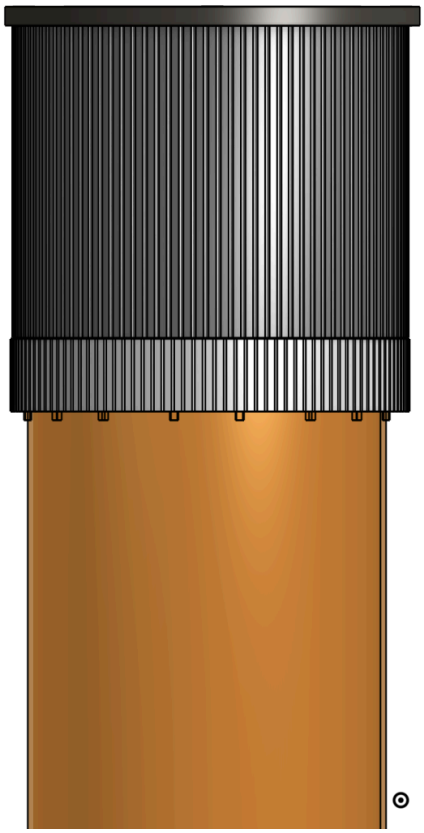

2
1
B
A
SCALE 1:2
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
OMER KHAWAR
CHECKED
KRISH PATEL
APPROVED
MATERIAL
FINISH
TITLE
PILL BOTTLE EXPLODED VIEW
SIZE
A
DWG NO.
REV
SCALE
1:1
SHEET
1 of 1

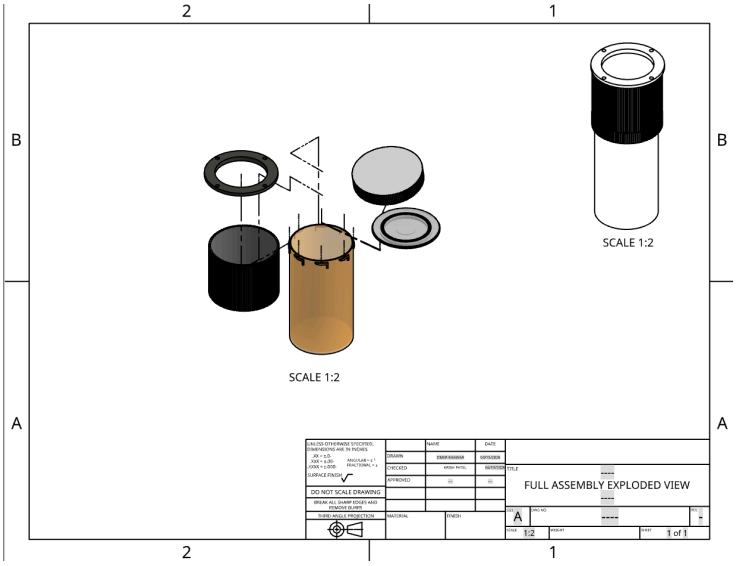







2
1
B
A
DO NOT SCALE DRAWING
CONTAINER EXPLODED VIEW




2
1
B
A
SCALE 1:2
SCALE 1:2
FULL ASSEMBLY EXPLODED VIEW
DO NOT SCALE DRAWING
A
1:2
1 of 1

